HOME LAB / PROJECT DOCUMENTATION

SysAdminLab

Windows Server and Active Directory

Complete project guide

39 documented milestones | 3 virtual machines

AD DS / DNS / DHCP / Group Policy / SMB
PowerShell account automation / Backup and recovery

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Core exercises complete. Client migration remains a follow-up.

A reproducible build record with exact commands, explanations, evidence labels, and a glossary. Passwords are excluded.

Long command lines may wrap visually in this reading copy. Use the editable Markdown guide or the original script files when copying commands.

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Project summary and results

This project simulates a small business network on one laptop. DC01 provides the domain directory, DNS, and Windows DHCP. FS01 provides IT and HR file shares and a separate virtual backup volume. CLIENT01 is the domain workstation used for standard-user testing.

Capability	Implemented result	Evidence
Virtual infrastructure	Three VMs on the SysAdminLab NAT Network, 10.10.10.0/24	Configuration screenshots and user reports, as identified in each milestone
Domain and DNS	corp.sampath.test; DC01 at 10.10.10.10	Initial DNS, service, and domain-controller checks screenshot-verified
Domain client and policy	CLIENT01 joined to CORP; LabComputers logon-message GPO applied	Domain sign-in reported; policy banner screenshot-verified
Group-based file access	IT and HR folders protected with NTFS and SMB permissions	Folder ACLs screenshot-verified; allowed and denied access passed by user report
Windows DHCP	DC01 distributes 10.10.10.100-199, with gateway .1 and DNS .10	CLIENT01 lease at 10.10.10.100 screenshot-verified
User provisioning	Alex Taylor and Maya Patel created from CSV into LabUsers and department groups	Script output and AD properties confirmed by user report
Account lifecycle	Alex disabled, fresh sign-in blocked, then re-enabled and sign-in restored	Passed by user report
File recovery	Recovery-Test.txt deleted and restored from backup version 10/25/2026-22:05	Catalog and original baseline screenshot-verified; restoration and matching hash passed by user report
Access after recovery	Alex reads the recovered original text; HR remains denied	Explicitly confirmed by user report
Final checkpoints	DC01-Backup-Recovery, FS01-Backup-Recovery, CLIENT01-Backup-Recovery	All three saved by user report

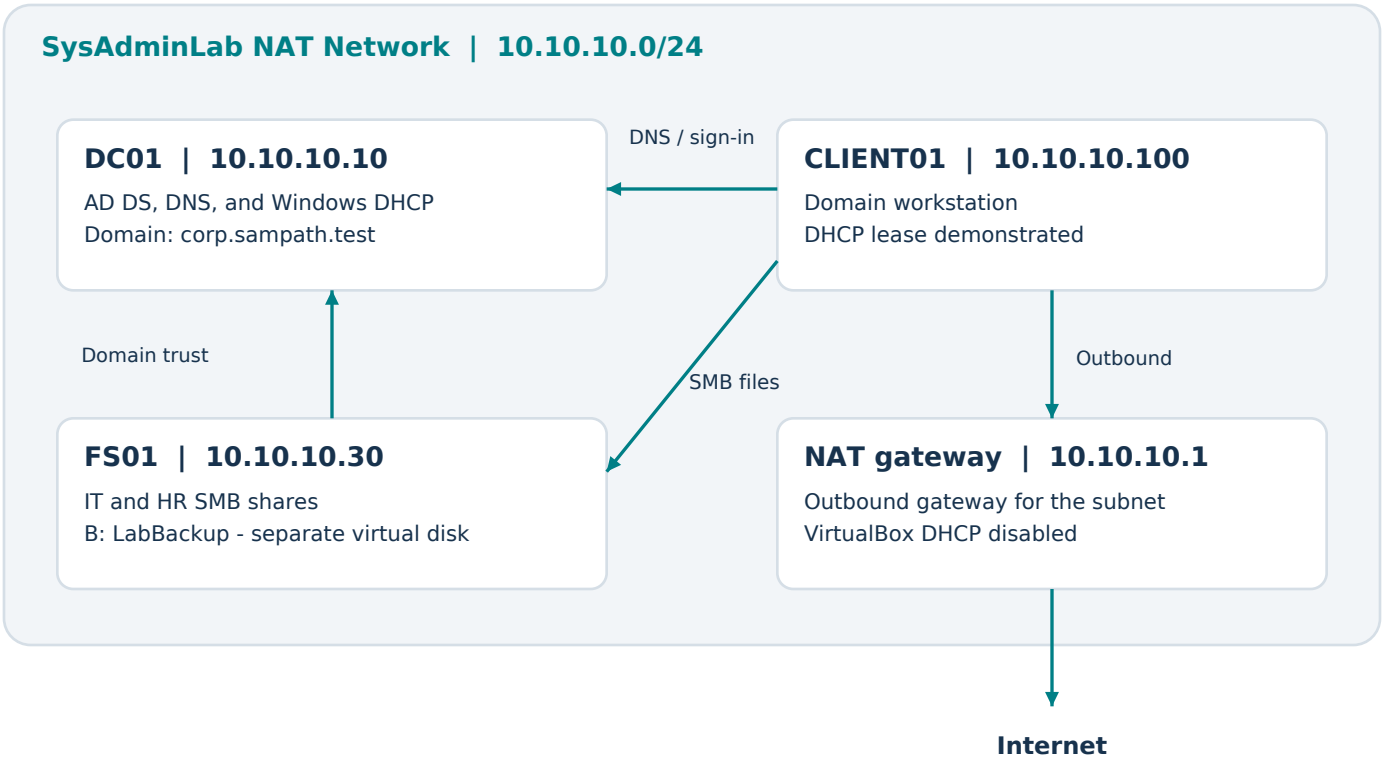
How to use this guide

The 39 milestones preserve the build sequence from VM installation through the final checkpoint. Every substantive command appears with its VM, shell, purpose, parameter meanings, and effect. Code blocks use PowerShell, Command Prompt, CSV, or plain text labels.

Read the evidence labels carefully: **screenshot-verified** means the displayed output was inspected; **user report** means Sampath confirmed the outcome without supplying that output. An **expected result** describes the target behavior. Planned or untested items are identified explicitly.

All Windows paths inside a VM refer to that VM's own drives. Host paths such as C:\SysAdminLab\ISO refer to the physical laptop. Passwords are excluded from this guide.

Lab topology



The diagram shows the main authentication, file-access, and outbound relationships. All three VMs share the same NAT Network and configured gateway. DC01 and FS01 keep static addresses; CLIENT01's demonstrated DHCP lease is 10.10.10.100 and can change in a future lease.

Final server and resource quick reference

Resource	Configuration
Domain / NetBIOS name	corp.sampath.test / CORP
Domain controller	DC01; Windows Server 2025 Standard Evaluation, Desktop Experience
File server	FS01; Windows Server Desktop Experience; setup used the server ISO, but FS01's final edition was not independently queried
FS01 system storage	Original 80 GB VDI; department folders on C:\Shares\IT and C:\Shares\HR
FS01 backup storage	Added 40 GB FS01-Backup.vdi; Disk 1 prepared as GPT; B: LabBackup, NTFS, Healthy by user report
DHCP scope	SysAdminLab-Clients; 10.10.10.100-10.10.10.199; active
DHCP options	Router 10.10.10.1; DNS 10.10.10.10; suffix corp.sampath.test
IT share / group	\\FS01\IT / CORP\GG_IT_Users
HR share / group	\\FS01\HR / CORP\GG_HR_Users
Script location	C:\LabScripts inside DC01
Recovery sample	C:\Shares\IT\Recovery-Test.txt inside FS01
Hash record	C:\Shares\Recovery-Test.sha256.txt inside FS01
Backup recovery version	10/25/2026-22:05, retained exactly from the catalog

Domain account	Department access	Functional test status
CORP\sampath	IT allowed; HR denied	Sign-in and IT file operations passed by user report
CORP\hr.test	HR allowed; IT denied	Sign-in and HR file operations passed by user report
CORP\alex.taylor	IT allowed; HR denied	Script provisioning, first sign-in/password change, disable/restore, and recovered-file read passed by user report
CORP\maya.patel	Member of GG_HR_Users	Stored AD properties and membership confirmed; first sign-in and share access not tested

The department groups are global security groups in LabGroups. The accounts are in LabUsers, and CLIENT01 is in LabComputers. The configured GPO is LabComputers-LogonMessage.

Environment and configuration record

Host and server

Item	Configuration
Host operating system	Windows 11 Home Single Language, version 25H2
Host processor	Intel Core i5-1240P
Host RAM	32 GB
Hardware virtualization	Enabled
Hypervisor	Oracle VirtualBox; Expert mode enabled
VM name and Windows computer name	DC01
Guest operating system	Windows Server 2025 Standard Evaluation - Desktop Experience
Guest RAM	4096 MB / 4 GB
Guest processors	2 virtual CPUs
Guest virtual disk	DC01.vdi; 80 GB capacity
Firmware	EFI enabled
VM folder	C:\SysAdminLab\DC01
Installation media	C:\SysAdminLab\ISO\Windows Server.iso
Baseline snapshot	DC01-Clean-Updated
Post-GPO snapshot	DC01-Domain-GPO; saved per user report

The ISO installed Windows Server 2025. VirtualBox's initially displayed Windows Server 2022 guest profile was a hardware preset.

First client

Item	Configuration and evidence
VM name / Windows computer name	CLIENT01; rename and hostname result reported by the user
Guest operating system	Windows 10 Pro; selected in Setup and desktop observed
Installation media	Existing Windows 10 ISO in C:\SysAdminLab\ISO; exact filename not recorded
VM folder	C:\SysAdminLab\CLIENT01
Virtual disk	C:\SysAdminLab\CLIENT01\CLIENT01.vdi; new 80 GB disk, full preallocation unchecked
Intended hardware allocation	4096 MB RAM, 2 virtual CPUs, EFI enabled; instructed values, final hardware screen not independently shown
Local account	LabAdmin; separate from domain accounts
Domain membership	corp.sampath.test; successful domain sign-in and verification results reported
Domain user	CORP\sampath / sampath@corp.sampath.test
Client snapshot	CLIENT01-Domain-GPO; saved per user report

Windows 10 Home and Pro reached the end of standard support on October 14, 2025. This existing ISO was chosen for the temporary lab client. Migration to a supported Windows 11 Pro release remains planned; no upgrade or Extended Security Updates enrollment is recorded.

Network

Setting	Value
VirtualBox attachment	NAT Network
Network name	SysAdminLab
IPv4 network prefix	10.10.10.0/24
VirtualBox DHCP	Disabled
NAT Network IPv6 option	Disabled for this IPv4 lab
Adapter 1	Enabled; virtual cable connected
DC01 IPv4 address	10.10.10.10
CLIENT01 IPv4 address	10.10.10.100 via DHCP from DC01; screenshot-verified. Initial static address was 10.10.10.20.
FS01 IPv4 address	10.10.10.30; static lab configuration, source address previously screenshot-verified
Windows DHCP server	DC01 at 10.10.10.10; CLIENT01's lease identifies this server
Subnet mask	255.255.255.0
Default gateway	10.10.10.1
Ethernet DNS client server after promotion	127.0.0.1 - DC01 itself
CLIENT01 DNS server	10.10.10.10 - DC01; screenshot-verified after enabling DHCP
Alternate DNS server	None configured

DC01's IP address, subnet mask, and gateway were verified in command output. Its preferred DNS client address was initially entered as 10.10.10.10. After promotion, PowerShell confirmed that Ethernet uses 127.0.0.1, the loopback address referring to DC01 itself. The internal DNS service was also queried successfully at 10.10.10.10. CLIENT01 now has the DHCP address 10.10.10.100 and uses DC01 at 10.10.10.10 for DNS. FS01 uses its configured static address 10.10.10.30. All three VMs are attached to SysAdminLab; the lab's configured mask is 255.255.255.0 and gateway is 10.10.10.1.

Phase 1 - Virtual machines and base networking

1. Build the Windows Server VM

Purpose: Create a server for learning without purchasing separate physical hardware.

Procedure:

1. Install VirtualBox on the physical laptop.
2. Download the official Windows Server 2025 Evaluation ISO and save it in C:\SysAdminLab\ISO.
3. Create DC01 using the resource settings above, attach the ISO, and use manual installation.
4. Install Windows Server 2025 Standard Evaluation with Desktop Experience.
5. Set the Administrator password locally and sign in.
6. In Server Manager → Local Server, change the Windows computer name to DC01 and restart the guest.

Why these choices: RAM supplies temporary working memory; virtual CPUs supply processing capacity; the VDI file provides the virtual hard disk. Desktop Experience provides graphical management windows. A clear server name helps identify the machine's intended job: DC01 means the first planned domain controller.

Verified result: Windows Server installed and Windows showed DC01 after the rename and restart. Domain-controller functionality is established later through promotion.

2. Patch the server and save a baseline

Purpose: Start with an updated server and preserve a rollback point before major configuration changes.

Procedure:

1. Install available Windows Server and security updates inside DC01.
2. Complete any requested restart.
3. Shut down DC01 normally.
4. In VirtualBox's Snapshots view, take a snapshot named DC01-Clean-Updated.

Verified result: Updates were reported complete, and the snapshot appeared in VirtualBox.

Recovery meaning: The snapshot captures the updated server before the shared network and static IP changes. Restoring it rolls back the VM's disk and saved settings to that point. A separate backup protects against losing the underlying VM files.

3. Create and attach the shared lab network

Purpose: Allow future lab VMs to communicate with DC01 while providing an outbound route through VirtualBox.

Procedure:

1. Enable Expert mode in VirtualBox Preferences to expose network-management tools.
2. Open Network Manager → NAT Networks.
3. Create SysAdminLab with prefix 10.10.10.0/24 and VirtualBox DHCP disabled.
4. With DC01 powered off, open Settings → Network → Adapter 1.
5. Select NAT Network, choose SysAdminLab, and keep the adapter and virtual cable enabled.
6. Save the settings, start DC01, and sign in.

Why NAT Network: Network Address Translation (NAT) supplies the outbound translation function. VirtualBox's shared NAT Network mode also allows attached VMs to communicate with one another.

Why VirtualBox DHCP is disabled: Dynamic Host Configuration Protocol (DHCP) automatically assigns addresses. VirtualBox DHCP was disabled so Windows Server could provide the service. At this early stage Windows DHCP was not configured; milestones 21-25 record its later installation and successful client lease.

Verified result: SysAdminLab was shown with prefix 10.10.10.0/24 and DHCP disabled. DC01's adapter was configured to use it.

4. Assign and verify a static server address

Purpose: Give DC01 a consistent network address that other lab computers can use.

Inside DC01, open Run and enter:

TEXT

```
ncpa.cpl
```

Command meaning and location: On DC01, ncpa.cpl is a Windows Run shortcut that opens Network Connections. It gives access to the Ethernet adapter's IP and DNS configuration.

Open Ethernet → Properties → Internet Protocol Version 4 (TCP/IPv4). Select manual IP and DNS settings and enter the IP address, mask, and gateway from the network table. For this initial step, set preferred DNS to 10.10.10.10, leave alternate DNS blank, and save. The effective DNS client setting after promotion is recorded in milestone 7.

Field	Practical meaning
IP address	Identifies DC01 on the network
Subnet mask	Determines which destinations are on the local subnet
Default gateway	Router used to reach destinations beyond the local subnet
DNS server	Resolves names and helps clients locate services, including a domain controller

The /24 prefix corresponds to subnet mask 255.255.255.0. The network identifier is 10.10.10.0; the ordinary host-address range is 10.10.10.1 through 10.10.10.254.

Verification commands:

Where and shell: Inside DC01, in Command Prompt. Both are diagnostic Windows utilities and also work in Windows PowerShell.

CMD

```
ipconfig
ping 10.10.10.1
```

Command	Meaning	Reason in this step
ipconfig	Displays basic network-adapter IP settings	Compare DC01's saved address, mask, and gateway with the lab plan
ping 10.10.10.1	Sends ICMP echo requests to the specified gateway address	Check whether DC01 can reach the virtual gateway

Verified output:

TEXT

```
IPv4 Address:     10.10.10.10
Subnet Mask:     255.255.255.0
Default Gateway: 10.10.10.1
```

Ping received four replies from 10.10.10.1.

What this confirms: The saved IPv4 settings are correct and DC01 can reach its virtual gateway. DNS resolution and domain services require separate checks.

DNS design: DC01 is configured to use itself because it will host the lab's Active Directory DNS service. Later, domain computers will use the internal DNS server, which can resolve external names through forwarders or root hints.

Phase 2 - Active Directory and DNS

5. Install and verify AD DS and DNS roles

Purpose: Add the server capabilities needed for central identity management and name resolution.

Procedure inside DC01:

1. Open Server Manager → Manage → Add Roles and Features.
2. Choose Role-based or feature-based installation and select DC01.
3. Select Active Directory Domain Services and DNS Server, accepting required management features.
4. Continue to confirmation and select Install.
5. Verify the two roles using Windows PowerShell:

POWERSHELL

```
Get-WindowsFeature AD-Domain-Services,DNS
```

Command explanation - DC01, Windows PowerShell: Get-WindowsFeature reads Windows Server role and feature installation status. AD-Domain-Services and DNS are the two feature names; the comma supplies both in one query. We run it after role installation to check that both required components are installed before promotion. This query displays status.

Verified result:

TEXT

```
[X] Active Directory Domain Services    AD-Domain-Services
[X] DNS Server                          DNS
```

The [X] markers confirm both roles are installed.

Role	Purpose
Active Directory Domain Services (AD DS)	Central directory for users, computers, and groups; supports domain authentication and management
Domain Name System (DNS)	Resolves names and service locations so clients can locate the domain controller and other resources

6. Create the forest and promote DC01

Purpose: Establish the lab's central directory so domain users, groups, computers, authentication, and policies can be managed together.

Procedure inside DC01:

1. Open Server Manager's notification flag and select Promote this server to a domain controller.
2. Select Add a new forest and enter corp.sampath.test as the root domain name. The .test suffix is reserved for testing.
3. On Domain Controller Options, select Windows Server 2025 for the forest and domain functional levels. Keep DNS Server and Global Catalog selected, and Read-only domain controller unselected.

4. Set and privately retain a strong Directory Services Restore Mode (DSRM) password. It is used for recovery; routine domain sign-in uses the Administrator account's existing password.
5. Leave Create DNS delegation unchecked for this new private lab domain. The warning about an unavailable authoritative parent DNS zone does not block this setup.
6. Use CORP as the NetBIOS domain name and keep the database, log, and SYSVOL paths below.
7. Continue through Review Options and Prerequisites Check. The checks passed, with the expected DNS delegation warning. Select Install and allow the automatic restart.
8. Sign in as CORP\Administrator using the existing Windows Server Administrator password. Open Server Manager → Tools → Active Directory Users and Computers.

Directory setting	Configured value
Forest root and domain DNS name	corp.sampath.test
NetBIOS domain name	CORP
First domain controller	DC01
Forest and domain functional levels selected in the wizard	Windows Server 2025
DNS Server / Global Catalog	Selected
Read-only domain controller	Unselected
AD DS database folder	C:\Windows\NTDS
Database log folder	C:\Windows\NTDS
SYSVOL folder	C:\Windows\SYSVOL

What the paths store: The AD DS database stores directory objects such as accounts, groups, and computers. Its transaction logs record database changes for recovery. SYSVOL stores shared Group Policy files and scripts.

Verified result: DC01 restarted after promotion, the sign-in screen displayed CORP\Administrator, and sign-in succeeded. Active Directory Users and Computers displayed corp.sampath.test. This confirms that the domain exists and the management console can access it. Basic DNS, service, and share checks were then completed in milestone 7.

Why a new forest: A forest is the top-level Active Directory structure. This lab starts with one forest, one domain, and one domain controller. The NetBIOS name CORP provides the short domain prefix in CORP\Administrator.

7. Verify the domain controller's basic services and DNS

Purpose: Check that DC01 publishes the names, services, and shared folders needed by domain computers before adding a client.

Run the following commands inside DC01 in Windows PowerShell:

POWERSHELL

```
Resolve-DnsName DC01.corp.sampath.test -Server 10.10.10.10 -Type A
Get-Service NTDS,DNS,Netlogon,DFSR
net share
Get-DnsClientServerAddress -AddressFamily IPv4
Resolve-DnsName _ldap._tcp.dc._msdcs.corp.sampath.test -Type SRV
Resolve-DnsName microsoft.com -Type A
```

Run each command separately so its output is easy to inspect.

Command meanings - all executed on DC01 in Windows PowerShell:

Command or parameter	Meaning and reason for using it here
Resolve-DnsName DC01.corp.sampath.test -Server 10.10.10.10 -Type A	Looks up DC01's IPv4 address by querying the DNS server explicitly at 10.10.10.10. Checks the server's own host record
Resolve-DnsName	Queries the supplied name; the following name is the target
-Server 10.10.10.10	Chooses the DNS server to query for this command
-Type A	Requests an IPv4 address record
Get-Service NTDS,DNS,Netlogon,DFSR	Reads the status of these four services. Commas select the service names; Running is the expected state
net share	Windows utility that lists this server's shared resources. Used to find SYSVOL and NETLOGON
Get-DnsClientServerAddress -AddressFamily IPv4	Reads configured IPv4 DNS-server addresses for local network interfaces. Checks which DNS server DC01's Ethernet adapter uses
-AddressFamily IPv4	Limits the displayed DNS-server addresses to the IPv4 family
Resolve-DnsName _ldap._tcp.dc._msdcs.corp.sampath.test -Type SRV	Queries the domain-controller service-location record. We check the returned target and port so clients can discover DC01
-Type SRV	Requests service-location records, which include a target hostname and port
Resolve-DnsName microsoft.com -Type A	Looks up a public hostname using the current resolver configuration. Checks external name resolution through DC01

These commands query information. The DNS tests check names and records; service status and share output provide separate checks of the server baseline.

Check	Verified result	What it confirms
DC01 host record	DC01.corp.sampath.test resolves to 10.10.10.10	The server's DNS name maps to its lab IPv4 address
Core services	NTDS, DNS, Netlogon, and DFSR all Running	The directory, DNS, domain logon, and SYSVOL replication services are running
Domain shares	SYSVOL and NETLOGON present	The shared locations for policy files and sign-in scripts are published

Check	Verified result	What it confirms
Ethernet DNS client setting	127.0.0.1	DC01 queries its own DNS service
Domain-controller service record	_ldap._tcp.dc._msdcs.corp.sampath.test returns dc01.corp.sampath.test, port 389, with address 10.10.10.10	The DNS record used to discover a domain controller is available
External DNS query	microsoft.com returns an IPv4 A record	External names can be resolved through DC01's current DNS configuration

The NETLOGON share points to C:\Windows\SYSVOL\sysvol\corp.sampath.test\SCRIPTS. SYSVOL points to C:\Windows\SYSVOL\sysvol.

Service meanings: NTDS is Active Directory Domain Services. DNS handles name resolution. Netlogon supports domain logon and registration of domain-controller DNS records. DFSR means Distributed File System Replication and is used to replicate SYSVOL between domain controllers; this lab currently has one domain controller.

Scope of verification: These checks establish the initial server baseline. Client domain joining and user sign-in were subsequently tested in milestones 9-10. The first Group Policy exercise was verified in milestone 11.

8. Save the configured domain-controller checkpoint

Purpose: Preserve the verified initial domain-controller configuration before adding the workstation.

1. Shut down DC01 normally from inside the guest, using the Windows shutdown controls or:

CMD

```
shutdown /s /t 0
```

Command meaning - DC01, Command Prompt or Windows PowerShell: shutdown is a Windows power-control utility. The /s option requests shutdown; /t 0 sets a zero-second delay. Its purpose here is to power off the guest before taking the snapshot. This changes the VM's power state. Windows shutdown controls were offered as an alternative; the exact shutdown method was not separately confirmed.

2. After VirtualBox shows Powered Off, open Snapshots → Take.
3. Name the snapshot DC01-AD-Configured.
4. Start DC01 again before testing the client.

Result: The user reported that this snapshot was saved. It captures the domain-controller baseline before the client join and the first standard user account.

Phase 3 - Domain client and Group Policy

9. Install, configure, and join CLIENT01

Purpose: Add a workstation to the company-style network so domain accounts can sign in to another computer.

VM installation:

1. In VirtualBox, create CLIENT01 in C:\SysAdminLab\CLIENT01, attach the existing Windows 10 ISO, and disable unattended installation.
2. Use the intended hardware allocation in the client table and create a new 80 GB CLIENT01.vdi with full preallocation unchecked.
3. Before starting Setup, attach Adapter 1 to NAT Network → SysAdminLab. Keep the adapter and virtual cable enabled.
4. In Windows Setup, select Windows 10 Pro, choose Custom installation, select the 80 GB unallocated drive, and allow Windows to create its partitions.
5. Complete the offline setup using Continue with limited setup. Create the local account LabAdmin and retain its password privately.
6. Inside Windows, run sysdm.cpl, open Computer Name → Change, set the name to CLIENT01, and restart.

Windows Run shortcut: sysdm.cpl opens System Properties on the VM where it is entered. Its Computer Name tab is used first to rename CLIENT01 and later to join the domain.

Static client network:

Open ncpa.cpl → Ethernet → Properties → Internet Protocol Version 4 (TCP/IPv4), and enter:

Field	Value
IP address	10.10.10.20
Subnet mask	255.255.255.0
Default gateway	10.10.10.1
Preferred DNS	10.10.10.10
Alternate DNS	Blank

The client uses DC01 for DNS so it can locate domain services. DHCP is still disabled, so the client receives its address through this manual configuration.

Connection test: With DC01 running, open PowerShell on CLIENT01 and run:

POWERSHELL

```
hostname
Test-NetConnection DC01.corp.sampath.test -Port 389
```

The user reported hostname CLIENT01, RemoteAddress 10.10.10.10, and TcpTestSucceeded True. This establishes name resolution and TCP connectivity to the directory service port; the domain join and sign-in provide the subsequent identity test.

Command meanings at this stage: hostname displays the local Windows computer name, so CLIENT01 is the expected result after the rename and restart. Test-NetConnection is a PowerShell diagnostic cmdlet; the supplied

server name is its destination and -Port 389 selects the TCP port to test. We use these checks before joining CLIENT01 to the domain to identify name or reachability problems early. The detailed parameter and output guide is repeated for the current FS01 exercise below.

Domain join:

1. On CLIENT01, run sysdm.cpl → Computer Name → Change.
2. Select Domain and enter corp.sampath.test.
3. When prompted, use CORP\Administrator and the existing DC01 Administrator password to authorize the join.
4. Accept the domain welcome and restart CLIENT01.
5. Use Other user on the sign-in screen for a domain account.

The domain Administrator credentials authorize the join. LabAdmin remains a local workstation account. The standard domain user created in milestone 10 is used for the workstation sign-in.

Evidence: The Windows 10 desktop, new virtual disk, and shared NAT Network attachment were observed in screenshots. The user reported saving the client settings, the successful connection test, and the later domain sign-in. No separate secure-channel test has been run.

10. Create the first standard user and verify domain sign-in

Purpose: Practice centralized account creation and show that a user stored on DC01 can sign in on CLIENT01.

On DC01, signed in as CORP\Administrator:

1. Open Server Manager → Tools → Active Directory Users and Computers.
2. Right-click corp.sampath.test → New → Organizational Unit.
3. Name it LabUsers and keep Protect container from accidental deletion selected.
4. Select LabUsers → New → User and enter the account details below.
5. Set a temporary password privately. Keep User must change password at next logon selected. Leave User cannot change password, Password never expires, and Account is disabled unselected, then finish account creation.

Account field	Value
First / last name	Sampath / Peddagolla
Displayed full name	Sampath Peddagolla
User logon name	sampath
User Principal Name (UPN)	sampath@corp.sampath.test
Short domain sign-in name	CORP\sampath
OU location	corp.sampath.test/LabUsers
Intended access	Standard domain user; no additional privileged membership configured

First sign-in on CLIENT01:

1. Select Other user.
2. Enter sampath@corp.sampath.test, or CORP\sampath, and the temporary password.

3. When required, enter the current temporary password in the old-password field and choose a different compliant password in both new-password fields.
4. Finish signing in to the desktop.

Password troubleshooting: Windows initially rejected a proposed password for length, complexity, or history requirements, then showed a credential error during a later attempt. The exact cause was not independently established. Guidance clarified the current temporary password versus a rejected proposed password and recommended a new password excluding the account and full-name tokens. The user subsequently reported a successful sign-in. No password-policy change or administrator password reset was recorded, and no passwords are stored in this guide.

For a future policy-related rejection, inspect the actual default domain policy on DC01 rather than guessing its configured minimum:

POWERSHELL

```
Get-ADDefaultDomainPasswordPolicy |
    Select-Object MinPasswordLength,ComplexityEnabled,PasswordHistoryCount,MinPasswordAge
```

This diagnostic command has not yet been run in the lab. The suggested 16-character password recipe was guidance, not a measured domain minimum.

Command explanation - DC01, Windows PowerShell; reference diagnostic only:

Part	Meaning and purpose
Get-ADDefaultDomainPasswordPolicy	Reads the default password policy for the signed-in domain; used to check the actual configured rules
Pipe symbol at the end of the first line	Sends the retrieved policy object into the next command
Select-Object	Chooses which properties to display from that object
MinPasswordLength	Minimum permitted password length in characters
ComplexityEnabled	Whether the domain's complexity requirement is enabled
PasswordHistoryCount	Number of previous passwords retained for reuse restrictions
MinPasswordAge	Minimum interval before a user can ordinarily change a password again

The two lines form one PowerShell pipeline. It reads and displays policy settings. Its output has not been captured, so this guide does not assign measured values to these rules.

Sign-in verification: On CLIENT01, open normal Command Prompt by running cmd and execute:

CMD

```
whoami
hostname
echo %LOGONSERVER%
```

Command	Expected result, confirmed by user report	Meaning
whoami	corp\sampath	The current session uses the domain account

Command	Expected result, confirmed by user report	Meaning
hostname	CLIENT01	The session is on the intended workstation
echo %LOGONSERVER%	\\DC01	The session identifies DC01 as its logon server

Where and purpose: These checks ran on CLIENT01 in Command Prompt after the standard user signed in. whoami identifies the current account; hostname identifies the workstation; echo prints the value of the LOGONSERVER environment variable. The percent-sign variable syntax belongs to Command Prompt. Together, the checks identify the user, computer, and logon server for the tested session. All three display information.

The user reported that all three outputs matched. No screenshot of this command output was supplied. The LabUsers OU and user-creation form were visually observed earlier.

Result: The initial two-computer domain works for the first standard-user sign-in. Computer organization and the first policy exercise follow in milestone 11. Group-based access remains to be configured.

11. Organize CLIENT01 and deploy the first Group Policy

Purpose: Organize workstation identities in a dedicated OU and demonstrate that a centrally managed computer setting reaches CLIENT01.

Create the computer OU and move the workstation account:

1. On DC01, open Server Manager → Tools → Active Directory Users and Computers.
2. Right-click corp.sampath.test → New → Organizational Unit.
3. Enter LabComputers, keep Protect container from accidental deletion selected, and save.
4. Open the domain's Computers container, right-click CLIENT01 → Move, and select LabComputers.
5. Open LabComputers and confirm CLIENT01 appears there.

Verified result: A screenshot showed the LabComputers OU selected and CLIENT01 listed with type Computer.

The computer account represents the workstation inside the directory. Moving it changes its directory location. An OU provides a place to link Group Policy for a chosen set of computers; LabUsers separately holds the first domain user.

Create and link the policy on DC01:

1. Open Server Manager → Tools → Group Policy Management.
2. Expand Forest: corp.sampath.test → Domains → corp.sampath.test.
3. Right-click LabComputers → Create a GPO in this domain, and Link it here.
4. Name the policy LabComputers-LogonMessage.
5. Expand LabComputers, right-click the policy, and select Edit.

The Group Policy Management Editor screenshot showed LabComputers-LogonMessage associated with DC01.CORP.SAMPATH.TEST.

Configure the sign-in message:

In the editor, navigate to Computer Configuration → Policies → Windows Settings → Security Settings → Local Policies → Security Options.

Double-click each of the following settings, select Define this policy setting, enter its value, and save with Apply → OK:

Policy setting	Value or reproduction example
Interactive logon: Message title for users attempting to log on	SysAdminLab
Interactive logon: Message text for users attempting to log on	Example: Welcome to SysAdminLab. Use your assigned domain account to sign in.

Computer Configuration is used because this message belongs to the workstation's sign-in process. The GPO is linked to LabComputers, which currently contains CLIENT01.

Apply and test on CLIENT01:

1. Keep DC01 running and switch to CLIENT01.
2. Open Command Prompt using Run as administrator. If prompted while signed in as the standard user, supply CLIENT01\LabAdmin and its privately retained local password.
3. Refresh the computer settings:

CMD

```
gpupdate /target:computer /force
```

Command explanation - CLIENT01, elevated Command Prompt: gpupdate is a Windows utility for refreshing Group Policy. The /target:computer option selects computer settings, and /force reapplies settings even when they have not changed. It was used after creating the logon-message GPO so CLIENT01 would process the policy before the restart test. This applies configuration to the computer; it is a configuration action rather than a status query.

4. Wait for the computer-policy update to finish successfully, then restart CLIENT01.
5. Observe the message before signing in.

The /target:computer option limits the refresh to computer settings. The /force option reapplies settings. LabAdmin authorizes the elevated local command; the workstation's domain identity is used for computer policy processing.

Verified result: The supplied CLIENT01 screenshot displayed the title SysAdminLab, a welcome message directing the user to sign in with the assigned domain account, and an OK button. This verifies the visible result of the first policy exercise. A separate screenshot of the gpupdate completion output was not provided.

What this demonstrates: The workstation is organized in Active Directory, the computer policy was created centrally on DC01, and the configured sign-in message is active on CLIENT01.

12. Save snapshots of the working domain and client

Purpose: Preserve the domain membership, first standard user, computer OU, and working sign-in-message policy before the department and file-server exercises.

1. Shut down CLIENT01 from inside Windows, then shut down DC01 normally.
2. Wait until both show Powered Off in VirtualBox.
3. For each VM, open Snapshots → Take and use the names below.
4. Start DC01 first, then CLIENT01.

VM	Snapshot name	Stage preserved
DC01	DC01-Domain-GPO	Domain, joined workstation account, first standard user, OUs, and first GPO
CLIENT01	CLIENT01-Domain-GPO	Joined client with the tested sign-in message

Result: The user reported both snapshots saved. These snapshots precede the creation of LabGroups and the HR test account.

Phase 4 - Department groups and file services

13. Create department security groups and the HR test account

Purpose: Establish department membership before assigning shared-folder permissions.

Create the groups on DC01:

1. Open Active Directory Users and Computers as CORP\Administrator.
2. Right-click corp.sampath.test → New → Organizational Unit.
3. Create LabGroups with accidental-deletion protection selected.
4. Right-click LabGroups → New → Group. Create each group below, choosing Group scope: Global and Group type: Security.

Group name	Department	Member established
GG_IT_Users	Information Technology	CORP\sampath
GG_HR_Users	Human Resources	CORP\hr.test

GG is the lab's naming prefix for Global Group. A global group collects accounts from the same domain. A security group can receive resource permissions, allowing access to be managed through membership.

Assign Sampath to IT:

1. Open LabGroups → GG_IT_Users → Properties → Members → Add.
2. Enter sampath, select Check Names, confirm Sampath Peddagolla resolves, and save with OK → Apply.

Create the HR test user:

1. Right-click LabUsers → New → User.
2. Enter first name HR, last name Test, and user logon name hr.test, retaining the suffix @corp.sampath.test.
3. Set a strong temporary password privately. Keep User must change password at next logon selected and the other account-creation boxes unselected.
4. Finish the wizard.
5. Open LabGroups → GG_HR_Users → Members → Add.
6. Enter hr.test, select Check Names, and save with OK → Apply.

HR account field	Configured value
Display name	HR Test
User Principal Name	hr.test@corp.sampath.test
Short sign-in name	CORP\hr.test
Directory location	corp.sampath.test/LabUsers
Department group	GG_HR_Users

Verified results: Screenshots showed both security groups in LabGroups, Sampath in GG_IT_Users' Members tab, and HR Test in GG_HR_Users' Members tab. Both member entries pointed to corp.sampath.test/LabUsers. Global scope was selected in the instructions; a separate General-tab screenshot of scope was not supplied.

Stage scope: HR Test's successful client sign-in and department access are recorded in milestone 20; the initial forced-password-change prompt was not separately reported. FS01 permissions are recorded in milestones 17-18. Sign out and sign back in after changing a user's group membership so a new session includes that membership.

14. Set up FS01 and verify its domain membership

Status: The user reports Windows Server setup complete after selecting the new 80 GB unallocated disk. A screenshot confirms hostname FS01, source address 10.10.10.30, and successful TCP connectivity to DC01 at 10.10.10.10:389. Following the domain-join and restart instructions, the user reported Test-ComputerSecureChannel returned True. The mask, gateway, and preferred-DNS values below were instructed and were not separately displayed in the screenshot.

VM build instructions used for this stage: Create a new FS01 VM in C:\SysAdminLab\FS01 using C:\SysAdminLab\ISO\Windows Server.iso, manual installation, a new 80 GB VDI, 4096 MB RAM, two virtual CPUs, and EFI enabled. Attach Adapter 1 to NAT Network → SysAdminLab. The instructed edition was Windows Server 2025 Standard Evaluation with Desktop Experience. Select the new unallocated disk during Setup, complete installation, and privately set the local Administrator password. Final edition and hardware-allocation screens were not independently captured; the graphical desktop and disk were observed.

Execution location: Windows PowerShell inside the FS01 guest VM. Keep DC01 running.

Intended FS01 setting	Value
Windows computer name	FS01
IPv4 address	10.10.10.30
Subnet mask	255.255.255.0
Default gateway	10.10.10.1
Preferred DNS	10.10.10.10
Alternate DNS	Blank

Set the computer name through sysdm.cpl → Computer Name → Change, then restart. Set the Ethernet IPv4 values through ncpa.cpl. These are Windows Run shortcuts, used here on FS01.

Clipboard integration during setup: To enable host-to-guest copying, eject the Windows installation ISO, choose VirtualBox Devices → Insert Guest Additions CD Image, and open the mounted CD inside the guest. Run VBoxWindowsAdditions.exe, install and restart, then choose Devices → Shared Clipboard → Host to Guest. The installer adds guest integration drivers and applications. The user reported completing this step while preparing FS01; completion on each of the other VMs was not separately confirmed.

Check 1: Identify the local computer

POWERSHELL

```
hostname
```

Meaning: hostname displays the Windows computer name. It is a Windows utility that works inside PowerShell.

Purpose and reason: Confirm that the guest was renamed to FS01 and that the command is being executed on the intended computer after its restart. The VirtualBox VM label and the guest's Windows computer name are separate settings.

Observed output from the FS01 screenshot:

TEXT

```
FS01
```

Effect and actual result: Displays information. The screenshot confirms the Windows computer name is FS01.

Check 2: Test the connection to the domain controller

POWERSHELL

```
Test-NetConnection DC01.corp.sampath.test -Port 389
```

Command part	Meaning
Test-NetConnection	PowerShell cmdlet that diagnoses a network connection
DC01.corp.sampath.test	Destination's fully qualified domain name: computer DC01 in corp.sampath.test
-Port	Parameter selecting the destination TCP port
389	LDAP port used by the domain controller

LDAP means Lightweight Directory Access Protocol. Active Directory uses LDAP for directory access. TCP means Transmission Control Protocol. A port number identifies a service endpoint on a computer.

Purpose and reason: Before attempting the domain join, check that FS01 resolves the intended server name and can establish a TCP connection to its LDAP endpoint. This helps locate network problems before working on domain membership.

Observed output from the FS01 screenshot:

TEXT

```
ComputerName     : DC01.corp.sampath.test
RemoteAddress    : 10.10.10.10
RemotePort       : 389
InterfaceAlias   : Ethernet
SourceAddress    : 10.10.10.30
TcpTestSucceeded : True
```

Output field	Meaning in this lab
RemoteAddress	Address reached for DC01; expected 10.10.10.10
RemotePort	Tested destination port; expected 389
InterfaceAlias	Local adapter used for the test
SourceAddress	FS01's address used for the connection; expected 10.10.10.30
TcpTestSucceeded	True indicates that the TCP connection test succeeded

How to interpret the result: The expected address and a successful TCP test establish this particular name-resolution and connection check. The subsequent secure-channel test below provides the member-server trust check. Active Directory also uses other services and ports, so port 389 alone is not a complete domain-health test.

If the name does not resolve, first review FS01's DNS setting and whether DC01 is running. If the name resolves but the TCP test fails, investigate the network path, service availability, and firewall rules before changing settings.

Effect and actual result: Performs diagnostic queries and a connection attempt without changing the configured network or joining the domain. The screenshot confirms the expected destination, source address, port, and TcpTestSucceeded True.

Join FS01 to the domain

Where and shell: Inside FS01, open Windows PowerShell using Run as administrator. Keep DC01 running.

POWERSHELL

```
Add-Computer -DomainName corp.sampath.test -Credential CORP\Administrator -Restart
```

Command part	Meaning and purpose in this step
Add-Computer	Joins this local computer, FS01, to the domain and creates its computer account if needed
-DomainName corp.sampath.test	Specifies the domain to join
-Credential CORP\Administrator	Uses the domain account authorized to perform the join; enter its password privately at the prompt
-Restart	Restarts FS01 after a successful join to complete the change

Reason: Make the file server a member of the same managed domain as the users and workstation, ready for domain-based access administration.

Effect and expected result: Changes domain membership and restarts FS01 after success. Following restart, sign in as FS01\Administrator with the local FS01 Administrator password.

Actual result: Following these join and restart instructions, the user reported the secure-channel check below returned True. This confirms the domain relationship by user report; the Add-Computer completion output was not independently captured.

After restart: Verify the member server's secure channel

Where and shell: Inside FS01 after joining and restarting, use Windows PowerShell opened with Run as administrator.

POWERSHELL

```
Test-ComputerSecureChannel
```

Meaning: Tests the trust channel between the local domain-member computer and its domain. The command uses the member computer's domain relationship.

Reason: Verify that FS01 can communicate through its domain trust before configuring shares.

Reported result on FS01:

TEXT

```
True
```

True indicates a working channel; False indicates a failed test. This is the diagnostic form with no repair parameter. This check is for FS01 as a member server; it is not the domain-controller check for DC01.

Actual result: The user reported True. This verifies a working domain secure channel for FS01 by user report. No screenshot of this particular output was supplied.

15. Install the File Server role on FS01

Purpose: Prepare the file-server role before creating IT and HR shared folders and assigning permissions. SMB means Server Message Block, the protocol CLIENT01 will use to access the shares.

Where and shell: Windows PowerShell opened with Run as administrator inside FS01.

POWERSHELL

```
Get-WindowsFeature FS-FileServer
```

Command part	Meaning and reason for use
Get-WindowsFeature	Reads Windows Server role and feature installation status on the local server
FS-FileServer	Internal role-service name for File Server; selects the component we need to check

Reason at this stage: Determine whether the File Server role service is already installed before preparing shares.

Output interpretation:

Result	Meaning
[X] File Server / Installed	Role service is already installed
[ ] File Server / Available	Role service is available to install

Effect and actual result: Reads status. The user reported Available, indicating that the role service is ready to install. No screenshot of this output was provided. File shares and department permissions have not yet been configured.

Install the File Server role service

Where and shell: Inside FS01, in Windows PowerShell opened with Run as administrator.

POWERSHELL

```
Install-WindowsFeature -Name FS-FileServer -IncludeManagementTools
```

Command part	Meaning
Install-WindowsFeature	Installs the selected Windows Server role service and its required dependencies
-Name FS-FileServer	Selects the File Server role service
-IncludeManagementTools	Includes applicable management tools

Reason: Enable FS01's File Server role service before preparing the IT and HR shared folders.

Effect: Changes the role configuration on FS01. Administrator elevation is required. The command targets the local server and does not request an automatic restart.

Expected result and interpretation:

Output	Meaning
Success: True	Installation succeeded
Exit Code: Success	Successful installation completion
Restart Needed: No	Continue with the next setup step
Restart Needed: Yes	Restart FS01 through Windows before continuing

Actual result: The user reported Exit Code Success and no restart required. This records successful installation by user report. No installation-output screenshot was supplied.

16. Create the department folders on FS01

Where and shell: Inside the FS01 VM, in Windows PowerShell opened with Run as administrator. The C: paths below belong to the virtual file server.

POWERSHELL

```
New-Item -ItemType Directory -Path "C:\Shares\IT","C:\Shares\HR" -Force
```

Command part	Meaning
New-Item	Creates items in the filesystem
-ItemType Directory	Creates folders
-Path	Specifies the full destination paths
"C:\Shares\IT","C:\Shares\HR"	Two folder paths; the comma supplies both to the command
-Force	Creates missing parent folders and allows existing directories to be returned

Reason at this stage: Give IT and HR separate folder locations so department permissions can be assigned independently before sharing them with CLIENT01.

Folder on FS01	Department group planned for access
C:\Shares\IT	CORP\GG_IT_Users
C:\Shares\HR	CORP\GG_HR_Users

Effect and expected result: Creates the local folders, including the parent C:\Shares if needed. Output should list IT and HR under C:\Shares. Department permissions and named SMB shares follow in later steps.

Actual result: The user reported completing the New-Item command for both folders. Folder creation is recorded as complete by user report; no output screenshot was supplied for independent review.

17. Configure and inspect department folder permissions

Where and shell: Inside the FS01 VM, in Windows PowerShell opened with Run as administrator. These commands target only the newly created IT and HR folders.

Purpose: Give each department's security group Modify permission on its own folder. Preserve Full control for the local Administrators group and Windows SYSTEM account. Remove inherited permission entries so broad permissions inherited from C:\Shares do not grant other ordinary users access.

NTFS means New Technology File System. NTFS permissions govern access to the folders and files on FS01. SMB share permissions will be configured in a later step; this step does not create network shares.

Assign the permissions

Run these commands one at a time:

POWERSHELL

```
icacls "C:\Shares\IT" /grant:r "Administrators:(OI)(CI)F" "SYSTEM:(OI)(CI)F"
  "CORP\GG_IT_Users:(OI)(CI)M" /inheritancelevel:r
icacls "C:\Shares\HR" /grant:r "Administrators:(OI)(CI)F" "SYSTEM:(OI)(CI)F"
  "CORP\GG_HR_Users:(OI)(CI)M" /inheritancelevel:r
```

Command part	Meaning and reason
icacls	Native Windows utility that displays or changes a file or folder's discretionary access control list; it can run from PowerShell
C:\Shares\IT / C:\Shares\HR	The specific folder whose permissions are being changed
/grant:r	Grants the listed permissions and replaces prior explicit grants for each listed identity; it does not erase unrelated explicit entries
Administrators:(OI)(CI)F	Gives the local Administrators group Full control on the department folder, with inheritance to files and subfolders
SYSTEM:(OI)(CI)F	Gives the Windows SYSTEM account Full control on the folder, with inheritance to files and subfolders
CORP\GG_IT_Users:(OI)(CI)M	Gives the IT department group Modify access to IT
CORP\GG_HR_Users:(OI)(CI)M	Gives the HR department group Modify access to HR
(OI)	Object inherit: files created beneath the folder inherit this permission
(CI)	Container inherit: subfolders inherit this permission
F	Full control, including the ability to change permissions
M	Modify: read, create, edit, and delete content; does not grant permission-management rights
/inheritancelevel:r	Disables inheritance from the parent folder and removes inherited permission entries, while retaining explicit entries
Quotes around each identity-and-permission argument	Keep the entire argument together so PowerShell passes its parentheses and punctuation to icacls correctly

Effect and expected result: Changes the two folders' access permissions. Each command should report Successfully processed 1 files; Failed processing 0 files. The word files is also used by icacls for a folder. The department groups receive Modify rather than Full control.

Inspect the resulting permissions

After both commands succeed, run:

POWERSHELL

```
icacls "C:\Shares\*"
```

Command part	Meaning
icacls with no permission-changing switches	Displays the existing access-control entries; it does not modify them
C:\Shares*	The wildcard selects the items immediately inside C:\Shares, which should be the IT and HR folders

Reason: Check the actual permissions before creating SMB shares and testing client access. IT should list Administrators and SYSTEM with F, plus CORP\GG_IT_Users with M. HR should list Administrators and SYSTEM with F, plus CORP\GG_HR_Users with M. Names may be displayed with prefixes such as BUILTIN\Administrators or NT AUTHORITY\SYSTEM. Unexpected extra entries or an error require review before proceeding.

Actual result: Verified in the user's screenshot image(20261025-034559).png. Each permission-setting command reported Successfully processed 1 files; Failed processing 0 files. The inspection reported Successfully processed 2 files; Failed processing 0 files. The displayed ACLs were:

Folder	Identity	Observed permissions
C:\Shares\IT	CORP\GG_IT_Users	(OI)(CI)(M)
C:\Shares\HR	CORP\GG_HR_Users	(OI)(CI)(M)
Both folders	NT AUTHORITY\SYSTEM	(OI)(CI)(F)
Both folders	BUILTIN\Administrators	(F), plus a separate (OI)(CI)(F) entry

The extra folder-only Administrators Full-control entry is redundant with its other Full-control entry and does not broaden department access. No broad Users or Everyone entry was displayed. This confirms folder configuration; network access and cross-department denial are still untested.

18. Create and inspect the IT and HR SMB shares

Where and shell: Inside FS01, in Windows PowerShell opened with Run as administrator.

Purpose: Publish the two existing department folders for CLIENT01 to reach over the lab network. SMB means Server Message Block. Department users receive Change permission at the share level; local administrators receive Full permission.

Run each creation command once:

POWERSHELL

```
New-SmbShare -Name "IT" -Path "C:\Shares\IT" -FullAccess "BUILTIN\Administrators" -ChangeAccess
  "CORP\GG_IT_Users"
New-SmbShare -Name "HR" -Path "C:\Shares\HR" -FullAccess "BUILTIN\Administrators" -ChangeAccess
  "CORP\GG_HR_Users"
```

Command part	Meaning
New-SmbShare	Creates a network share for an existing folder
-Name IT / -Name HR	Sets the share name used after the server name in a network path
-Path	Selects the existing local department folder on FS01
-FullAccess BUILTIN\Administrators	Grants the local administrators Full share access
-ChangeAccess CORP\GG_IT_Users / CORP\GG_HR_Users	Grants the matching department group Change share access for reading, creating, editing, and deleting content

Effect and expected result: Changes FS01's share configuration and returns information about each new share. IT should map to C:\Shares\IT; HR should map to C:\Shares\HR. The intended network paths are \\FS01\IT and \\FS01\HR. Existing NTFS permissions still apply.

Inspect both share ACLs:

POWERSHELL

```
Get-SmbShareAccess -Name "IT","HR"
```

Command part	Meaning and reason
Get-SmbShareAccess	Reads the share's access-control entries so the granted identities and rights can be checked
-Name	Chooses the share names to inspect
"IT","HR"	A comma-separated array selecting both department shares

Effect: Read-only; this inspection does not change permissions.

Share	AccountName expected	AccessControlType expected	AccessRight expected
IT	BUILTIN\Administrators	Allow	Full
IT	CORP\GG_IT_Users	Allow	Change
HR	BUILTIN\Administrators	Allow	Full
HR	CORP\GG_HR_Users	Allow	Change

Actual result: The user reported that the result matched the expected share-access entries after running the creation and inspection instructions. Share creation and the expected Administrators Full / matching department Change entries are recorded as complete by user report, without an output screenshot. This is configuration evidence; client access is tested separately below.

19. Test IT-user access from CLIENT01

Where: Inside CLIENT01. Keep DC01 and FS01 running. Use the standard CORP\sampath account for the test.

Purpose: Demonstrate that a department user can work with files in the allowed share and is denied access to the other department. A successful file operation verifies practical access beyond the server's permission listings.

Refresh the sign-in session and verify the account

1. On CLIENT01, use Start, the user/account icon, and Sign out. Sign in again as CORP\sampath. A full sign-out and sign-in refreshes group membership in the user's session.
2. Open Command Prompt on CLIENT01 through Windows+R, type cmd, and press Enter. Run:

CMD

```
whoami
```

Command part	Meaning
whoami	Displays the account running this Command Prompt

Reason and effect: Read-only identity check. Expected output is corp\sampath; capitalization may differ. Testing with the standard user makes the result meaningful because an administrator has access to both departments.

Test the allowed IT share

On CLIENT01, press Windows+R and enter this network path:

TEXT

```
\\FS01\IT
```

Path part	Meaning
\\	Begins a UNC (Universal Naming Convention) network path
FS01	The server hosting the share
IT	The SMB share name, mapped to C:\Shares\IT on FS01

In File Explorer, create a text file such as IT-Test.txt, enter a sentence, and save it. Reopen it, edit and save the content, then delete that test file.

Effect and expected result: The path opens the network share. The file actions change only the disposable test file on FS01 through CLIENT01. Opening, creating, reading, editing, and deleting should all succeed for CORP\sampath.

Test denied HR access with the same account

Remain signed in as CORP\sampath. Press Windows+R and enter:

TEXT

```
\\FS01\HR
```

The path syntax is the same; HR selects the share mapped to C:\Shares\HR on FS01. Expected result: Windows reports Access denied or that the user does not have permission. A credential prompt is not a confirmed denial result; cancel and report it so that the test can be checked without switching identities.

Test	Account	Expected result	Actual result
Identity check on CLIENT01	CORP\sampath	whoami shows corp\sampath	Included in procedure; output not separately supplied
IT share: open, create, read, edit, delete test file	CORP\sampath	Allowed	Passed by user report
HR share: open	CORP\sampath	Denied	Passed by user report

Actual result: When asked whether the IT file test passed and HR access was denied, the user answered yes. Both outcomes are recorded as passed by user report, without screenshots or a separate whoami output. These results demonstrate permitted IT file operations and denied cross-department HR access in the instructed standard-user session.

20. Test HR-user sign-in and reciprocal access

Where: Inside CLIENT01; keep DC01 and FS01 running.

Purpose: Test the second department with its own standard account. This checks that the HR share is usable by its intended group and that IT access is denied to HR.

1. Sign out of CLIENT01 using Start, the account icon, and Sign out. Sign in as CORP\hr.test with the temporary password set when the HR Test user was created. If prompted to change the password at first sign-in, set a new password privately.
2. Open a normal Command Prompt on CLIENT01 using Windows+R and cmd, then run:

CMD

```
whoami
```

Command part	Meaning and reason
whoami	Reads the current account name; expected output is corp\hr.test, confirming this test uses the HR domain account

Effect: Read-only identity query; it does not change the account or permissions.

3. Use Windows+R to open each network path below while signed in as HR Test:

UNC path on CLIENT01	Action and expected result
\\FS01\HR	Folder opens. Create a disposable text file such as HR-Test.txt, write and save text, reopen and edit it, then delete it. All file operations should succeed.
\\FS01\IT	Opening the share is denied; Windows should report Access denied or no permission.

These are UNC (Universal Naming Convention) paths: FS01 identifies the file server and HR or IT identifies the share. Opening a path through Run launches File Explorer; these paths are not PowerShell commands. Creating, editing, and deleting the test file changes that file on FS01 through the network.

A credential prompt is not a confirmed access-denial result. Cancel and report it instead of supplying another account for the test.

Test	Expected	Actual
HR Test's first domain sign-in	Successful; first-sign-in password change completed if requested	HR-account session confirmed by user report; password-change prompt not separately reported
whoami on CLIENT01	corp\hr.test	Included in procedure; output not separately supplied
HR share file operations as HR Test	Allowed	Passed by user report
IT share access as HR Test	Denied	Passed by user report

Actual result: The user explicitly confirmed that HR file operations worked and only IT access was denied while following the HR-account test instructions. Both reciprocal access checks are recorded as passed by user report, without screenshots or a separate whoami output. The first-sign-in password-change prompt was not separately described, and no password is recorded here.

The department-access exercise is complete by user report: CORP\sampath has working IT access and denied HR access; CORP\hr.test has working HR access and denied IT access.

Historical file-sharing snapshot plan

DC01-FileShares, FS01-FileShares, and CLIENT01-FileShares were suggested after the first share tests. Their creation was not confirmed. They are not listed as completed snapshots. The later three-VM Backup-Recovery checkpoint in milestone 39 was explicitly confirmed by the user.

Phase 5 - Windows DHCP

21. Install the DHCP Server role on DC01

Where and shell: Inside DC01, signed in as CORP\Administrator, in Windows PowerShell opened with Run as administrator.

Purpose: Add DHCP (Dynamic Host Configuration Protocol) services to DC01 for this lab. DHCP will later assign an IP address, subnet mask, gateway, and DNS settings automatically to CLIENT01. DC01 already has the static address 10.10.10.10.

POWERSHELL

```
Install-WindowsFeature -Name DHCP -IncludeManagementTools
```

Command part	Meaning and reason
Install-WindowsFeature	Installs the selected Windows Server role and required components on the local server
-Name DHCP	Selects the DHCP Server role
-IncludeManagementTools	Includes the relevant management tools, such as the DHCP console and PowerShell management module

Effect: Changes DC01's installed server-role configuration. Run locally in DC01's elevated Windows PowerShell. Keep the existing manual IPv4 settings on all three VMs at this stage.

Output field	Expected result
Success	True
Restart Needed	No
Exit Code	Success

Reason for the sequence: Install the role first; authorize DC01 in Active Directory and configure its DHCP address pool and network options afterward. CLIENT01 can then be switched to automatic addressing and its lease verified. DC01 and FS01 will retain their static server addresses.

Actual result: The user reported matching the expected installation result: Success True, RestartNeeded No, and ExitCode Success. Installation is recorded as successful by user report; no output screenshot was supplied. Authorization and address-pool setup are documented in the following steps.

22. Authorize DC01 as a DHCP server in Active Directory

Where and shell: Inside DC01, signed in as CORP\Administrator, in the elevated Windows PowerShell window used for role installation.

Purpose: Register DC01 in Active Directory's authorized DHCP-server list before configuring address assignment.

Register the server

POWERSHELL

```
Add-DhcpServerInDC -DnsName "DC01.corp.sampath.test" -IPAddress 10.10.10.10
```

Command part	Meaning
Add-DhcpServerInDC	Adds a DHCP server to the authorized-server list in Active Directory and triggers its authorization check
-DnsName	Identifies the DHCP server by its full DNS hostname
DC01.corp.sampath.test	The lab server being authorized
-IPAddress 10.10.10.10	Registers DC01's existing static address, rather than a client lease address

Effect: Changes the authorization entries stored in AD. The command normally returns to the prompt without output. Use the following query to check the resulting entry.

Check the authorization entry

POWERSHELL

```
Get-DhcpServerInDC
```

Command part	Meaning and reason
Get-DhcpServerInDC	Reads Active Directory's authorized DHCP-server list so the server name and IP can be checked

Effect: Read-only directory query.

IPAddress expected	DnsName expected
10.10.10.10	DC01.corp.sampath.test

Hostname capitalization may differ. This verifies the authorization record; address-pool configuration and client leasing will be checked in later steps.

Actual result: The user confirmed that Get-DhcpServerInDC matched the expected entry: 10.10.10.10 and DC01.corp.sampath.test. The authorization record is confirmed by user report, without a screenshot. The successful client lease is subsequently demonstrated in milestone 24.

23. Create the client DHCP scope and network options

Where and shell: Inside DC01, signed in as CORP\Administrator, in Windows PowerShell opened with Run as administrator.

Purpose: Prepare a pool of client addresses on the existing SysAdminLab network and specify the network settings that DHCP clients will receive. A scope defines the pool and subnet; a lease is a temporary IP-address assignment to a client.

Scope setting in the instructions	Value	Reason in this lab
Name	SysAdminLab-Clients	Identifies the client pool
Scope ID / subnet	10.10.10.0/24	The existing shared NAT Network

Scope setting in the instructions	Value	Reason in this lab
Start and end	10.10.10.100 through 10.10.10.199	100 client addresses; existing static addresses are outside this range
Subnet mask	255.255.255.0	Matches the lab subnet
State	Active	Enables the scope to respond to client requests
Router / default gateway	10.10.10.1	VirtualBox NAT gateway
DNS server	10.10.10.10	DC01 resolves the AD domain and other names
DNS domain	corp.sampath.test	Connection-specific DNS suffix supplied to clients

VirtualBox's built-in DHCP was disabled earlier for SysAdminLab. DC01 and FS01 retain their static addresses; CLIENT01's switch to automatic addressing follows server-side configuration and inspection.

Create the address pool

POWERSHELL

```
Add-DhcpServerv4Scope -Name "SysAdminLab-Clients" -StartRange 10.10.10.100 -EndRange 10.10.10.199
  -SubnetMask 255.255.255.0 -State Active
```

Command part	Meaning
Add-DhcpServerv4Scope	Creates an IPv4 DHCP scope on the local server
-Name	Sets its descriptive label
-StartRange / -EndRange	Sets the first and last client addresses, inclusive
-SubnetMask	Sets the subnet mask
-State Active	Enables the scope

Effect and reason: Changes DC01's DHCP configuration. The selected pool is separate from the existing server and gateway addresses.

Set gateway and DNS options

POWERSHELL

```
Set-DhcpServerv4OptionValue -ScopeId 10.10.10.0 -Router 10.10.10.1 -DnsServer 10.10.10.10 -DnsDomain
  "corp.sampath.test"
```

Command part	Meaning
Set-DhcpServerv4OptionValue	Sets DHCPv4 options to give clients
-ScopeId 10.10.10.0	Applies these options to this subnet's scope

Command part	Meaning
-Router 10.10.10.1	Supplies the default gateway
-DnsServer 10.10.10.10	Supplies DC01 as the DNS server
-DnsDomain corp.sampath.test	Supplies the DNS suffix

Effect and reason: Changes the selected scope's options so clients receive domain-aware name resolution and a route beyond the local subnet. The scope ID is its network identifier, not DC01's host address.

Inspect the pool and options

Run these read-only commands after both configuration commands complete:

POWERSHELL

```
Get-DhcpServerv4Scope -ScopeId 10.10.10.0
Get-DhcpServerv4OptionValue -ScopeId 10.10.10.0
```

Command part	Meaning and reason
Get-DhcpServerv4Scope	Reads scope settings so the range, mask, name, and active state can be checked
Get-DhcpServerv4OptionValue	Reads the configured scope options so gateway and DNS values can be checked
-ScopeId 10.10.10.0	Selects the lab's client scope for either query

Expected result: The scope query should show SysAdminLab-Clients, ScopeId 10.10.10.0, State Active, StartRange 10.10.10.100, EndRange 10.10.10.199, and mask 255.255.255.0. The options query should include router 10.10.10.1, DNS server 10.10.10.10, and DNS domain corp.sampath.test. Configuration commands may return to the prompt without output; these queries inspect the saved settings.

Actual result: The user reported completing the step after the four commands were provided. Step completion is recorded by user report; the full scope and option-query outputs were not supplied. The subsequent CLIENT01 screenshot verifies an assigned address within this pool and the intended gateway, DNS server, and DNS suffix, as recorded in milestone 24.

24. Switch CLIENT01 to DHCP and inspect its lease

Where: Inside CLIENT01. Keep DC01 and FS01 running. Sign out of the current standard-user session and sign in as CLIENT01\LabAdmin, the client VM's local administrator, to change the adapter settings.

Purpose: Demonstrate that DC01 supplies a client lease and network settings automatically. This change applies to CLIENT01's Ethernet adapter. DC01 and FS01 continue using their existing static server addresses.

Enable automatic IP and DNS settings

Press Windows+R inside CLIENT01 and enter:

TEXT

```
ncpa.cpl
```

Command part	Meaning and reason
ncpa.cpl	Opens Network Connections, where CLIENT01's Ethernet adapter settings can be changed

This opens a Control Panel applet; running the shortcut alone does not change network configuration.

1. Right-click Ethernet and select Properties.
2. Select Internet Protocol Version 4 (TCP/IPv4), then Properties.
3. Select Obtain an IP address automatically.
4. Select Obtain DNS server address automatically.
5. Select OK and close the adapter properties.

Effect: Replaces CLIENT01's manual IPv4 and DNS configuration with DHCP-provided settings. Its previous static address was 10.10.10.20; the expected DHCP address will be in the configured client pool.

Request a lease and inspect the assigned settings

Inside CLIENT01, open Command Prompt with Run as administrator and run these commands one at a time:

CMD

```
ipconfig /renew
ipconfig /all
```

Command part	Meaning and reason
ipconfig	Windows utility for TCP/IP configuration
/renew	Requests or renews DHCP configuration on DHCP-enabled adapters
/all	Displays full adapter settings, including the DHCP server, DNS servers, and lease information

Effect: Renewal can update the assigned network configuration. The /all query is read-only. Inspect the Ethernet adapter section.

Ethernet field	Expected result
DHCP Enabled	Yes
IPv4 Address	An address from 10.10.10.100 through 10.10.10.199
Subnet Mask	255.255.255.0
Default Gateway	10.10.10.1
DHCP Server	10.10.10.10
DNS Servers	10.10.10.10
Connection-specific DNS Suffix	corp.sampath.test
Lease Obtained / Lease Expires	Assigned lease timestamps; exact times depend on execution

The DHCP Server field identifies which server supplied the lease. Checking it alongside the address range and DNS values confirms that the client obtained the intended configuration from DC01.

Actual result: Verified in the user's screenshot image(20261025-194537).png. Host Name is CLIENT01. Ethernet shows DHCP Enabled Yes, IPv4 10.10.10.100 (Preferred), mask 255.255.255.0, gateway 10.10.10.1, DHCP Server 10.10.10.10, DNS Servers 10.10.10.10, and connection-specific DNS suffix corp.sampath.test. Lease Obtained and Lease Expires timestamps are present. This verifies a DHCP lease with the intended client configuration; a separate /renew completion message was not shown. Continued domain and share access after the address change is checked next.

25. Confirm the server lease and continued domain/share access

Purpose: Match the client's observed lease to DC01's lease records, check CLIENT01's machine trust, and confirm a standard domain user can still use the file share after the addressing change.

Check the lease record on DC01

Where and shell: Inside DC01, as CORP\Administrator, in Windows PowerShell opened with Run as administrator.

POWERSHELL

```
Get-DhcpServerv4Lease -ScopeId 10.10.10.0
```

Command part	Meaning and reason
Get-DhcpServerv4Lease	Reads active DHCPv4 leases so the server's record can be matched to CLIENT01
-ScopeId 10.10.10.0	Selects the lab's client-address scope

Effect and expected result: Read-only query. Look for 10.10.10.100, a hostname identifying CLIENT01, and AddressState Active. Hostname capitalization or a domain suffix may vary.

Check CLIENT01's domain trust

Where and shell: Inside CLIENT01, in the CLIENT01\LabAdmin session, open Windows PowerShell with Run as administrator.

POWERSHELL

```
Test-ComputerSecureChannel
```

Command part	Meaning and reason
Test-ComputerSecureChannel	Tests the local domain member's secure channel to its domain using Netlogon

Effect and expected result: Tests the trust without repairing it; expected output is True. This check is applied to CLIENT01, a domain member. It verifies that machine trust still works after the client acquired its DHCP settings.

Check the standard user's file access

On CLIENT01, sign out of LabAdmin and sign in as CORP\sampath. Press Windows+R and open:

TEXT

```
\\FS01\IT
```

This UNC path identifies the IT share on FS01. Create a disposable text file such as DHCP-Test.txt, enter text and save it, then reopen and delete the test file.

Effect and expected result: Changes only the disposable test file on FS01. Successful domain sign-in and file operations demonstrate that the standard user still reaches the permitted share after the client address changed.

Check	Expected	Actual
DC01 lease record	CLIENT01 / 10.10.10.100 / Active	Passed by user report
Secure-channel test on CLIENT01	True	Passed by user report
CORP\sampath sign-in and IT share file operations	Successful	Passed by user report

Actual result: The user confirmed that all three final checks passed. DC01's matching active client lease, CLIENT01's True secure-channel result, and successful CORP\sampath IT-share file operations are recorded by user report; no additional screenshots or command outputs were supplied for these checks. Together with the client lease screenshot, these results complete the planned DHCP exercise.

Phase 6 - PowerShell account automation

26. Prepare CSV input for account-management automation

Where and shell: Inside DC01, signed in as CORP\Administrator, in Windows PowerShell opened with Run as administrator. Notepad is used to save the input file on DC01's virtual C: drive.

Purpose: Prepare structured input for a script that will create two fictional domain users, place them in LabUsers, and add them to the appropriate existing department group. Account creation itself follows file preparation and inspection.

Create the working folder

POWERSHELL

```
New-Item -ItemType Directory -Path "C:\LabScripts" -Force
```

Command part	Meaning and reason
New-Item	Creates an item in the filesystem
-ItemType Directory	Makes a folder
-Path C:\LabScripts	Selects the folder inside DC01 where input and scripts will be kept
-Force	Creates missing parents and allows an already existing directory to be returned

Effect and expected result: Creates or returns the LabScripts folder. For a directory, this command preserves existing contents. It changes the local filesystem only.

Create the CSV file in Notepad

CSV means Comma-Separated Values. The first line names the columns; each following line supplies one user's details. Open Notepad on DC01 and enter:

CSV

```
FirstName,LastName,Username,Department
Alex,Taylor,alex.taylor,IT
Maya,Patel,maya.patel,HR
```

Column	Meaning
FirstName	User's given name
LastName	User's surname
Username	Planned domain sign-in name
Department	IT or HR; the creation script will select the corresponding security group

Use File, Save As with these values:

Save field	Value
Full file path	C:\LabScripts\NewUsers.csv

Save field	Value
Save as type	All Files (*.*)
Encoding	UTF-8

The planned user-to-group mapping is:

Planned account	Department	Planned security group
CORP\alex.taylor	IT	GG_IT_Users
CORP\maya.patel	HR	GG_HR_Users

A temporary password will be entered interactively during the later account-creation step.

Import and inspect the two records

Return to DC01's elevated Windows PowerShell and run:

POWERSHELL

```
Import-Csv -Path "C:\LabScripts\NewUsers.csv"
```

Command part	Meaning and reason
Import-Csv	Reads CSV rows into PowerShell objects; the headings become property names
-Path	Selects the input file to read

Effect and expected result: Reads the file so the headings and values can be checked before provisioning. Expected output contains two records: Alex Taylor / alex.taylor / IT and Maya Patel / maya.patel / HR. This inspection leaves AD accounts unchanged.

Actual result: The user confirmed the CSV preparation and inspection step. Folder/file preparation and inspection are recorded by user report; the actual Import-Csv output was not supplied. Account creation is performed by the next script.

27. Create domain users and department memberships from CSV

Where and shell: Inside DC01, signed in as CORP\Administrator. Save the script in a new Notepad window, then run it in Windows PowerShell opened with Run as administrator.

Purpose: Read each CSV record, create an enabled user in LabUsers, require a password change at first sign-in, and add the user to the existing department group. This is the first account-provisioning automation exercise.

Save the script

Use File, Save As in a new Notepad window. File name: C:\LabScripts\New-LabUsers.ps1. Save as type: All Files (*.*). Encoding: UTF-8. A .ps1 file contains a PowerShell script.

POWERSHELL

```
$ErrorActionPreference = "Stop"
Import-Module ActiveDirectory

$OU = "OU=LabUsers,DC=corp,DC=sampath,DC=test"
$Groups = @{
    IT = "GG_IT_Users"
    HR = "GG_HR_Users"
}
$Users = Import-Csv -Path "C:\LabScripts\NewUsers.csv"

$null = Get-ADOrganizationalUnit -Identity $OU
foreach ($Group in $Groups.Values) {
    $null = Get-ADGroup -Identity $Group
}

foreach ($User in $Users) {
    $Username = $User.Username.Trim()
    $Department = $User.Department.Trim()

    if ($Username -notmatch '^[a-z][a-z0-9._-]{0,19}$') {
        throw "Invalid username: $Username"
    }
    if (-not $Groups.ContainsKey($Department)) {
        throw "Unknown department: $Department"
    }
    if (Get-ADUser -Filter "SamAccountName -eq '$Username'") {
        Write-Host "SKIPPED: $Username already exists"
        continue
    }

    $GroupName = $Groups[$Department]
    $Password = Read-Host "Temporary password for $Username" -AsSecureString
    $FullName = "$($User.FirstName.Trim()) $($User.LastName.Trim())"
    $Parameters = @{
        Name = $FullName
        DisplayName = $FullName
        GivenName = $User.FirstName.Trim()
        Surname = $User.LastName.Trim()
        SamAccountName = $Username
        UserPrincipalName = "$Username@corp.sampath.test"
        Department = $Department
        Path = $OU
        AccountPassword = $Password
        Enabled = $true
        ChangePasswordAtLogon = $true
    }

    try {
        New-ADUser @Parameters
        Add-ADGroupMember -Identity $GroupName -Members $Username
        Write-Host "CREATED: $Username -> $GroupName"
    }
    catch {
        Write-Warning "FAILED: $Username - $($_.Exception.Message)"
    }
}
```

Script commands: meanings, reasons, and effects

All commands in this script run on DC01 as part of the script invocation below.

Command or setting	Meaning and reason	Effect
$ErrorActionPreference = Stop	Makes command errors terminating so they can be caught or stop processing	Sets script error behavior
Import-Module ActiveDirectory	Loads the AD management commands used below	Changes the current PowerShell session
Import-Csv -Path	Reads the prepared NewUsers.csv input	Read-only file access
Get-ADOrganizationalUnit -Identity $OU	Checks that the target LabUsers OU exists, using its distinguished name	Read-only AD query
Get-ADGroup -Identity $Group	Checks that both configured department groups exist before user creation begins	Read-only AD query
Get-ADUser -Filter	Searches for the CSV username so an existing account is skipped	Read-only AD query
Read-Host ... -AsSecureString	Prompts separately for each new user's temporary password, masks input, and returns a SecureString	Reads private console input
New-ADUser @Parameters	Creates the account using the named values in the Parameters hashtable	Changes AD user objects
Add-ADGroupMember -Identity $GroupName -Members $Username	Adds the newly created username to the mapped IT or HR group	Changes AD group membership
Write-Host	Displays CREATED or SKIPPED status	Console output
Write-Warning	Displays FAILED with the caught error message	Console output

Values supplied to New-ADUser

Hashtable key / parameter	Meaning in this script
Name	Full name used for the new directory object
DisplayName	Full name displayed for the person
GivenName / Surname	First and last names from the CSV
SamAccountName	Short domain sign-in name, such as alex.taylor
UserPrincipalName	Username followed by @corp.sampath.test
Department	IT or HR, read from the CSV
Path	OU=LabUsers,DC=corp,DC=sampath,DC=test
AccountPassword	SecureString read at the console prompt
Enabled = $true	Enables the new account for sign-in
ChangePasswordAtLogon = $true	Requires the user to change the temporary password at first sign-in

PowerShell constructs used

Construct	Meaning and purpose
$variable = value	Stores settings or a value for later use
@{ ... }	Hashtable; used for department-to-group mapping and named New-ADUser parameters
@Parameters	Splatting: passes the hashtable entries as named command parameters
foreach	Repeats group checks or account processing for each item
$User.FirstName and similar properties	Reads the corresponding CSV column
.Trim()	Removes surrounding whitespace from imported field values
$Groups.Values	Enumerates the two configured group names for preflight checks
.ContainsKey() and $Groups[$Department]	Validates the department and retrieves its group name
-notmatch and the username pattern	Accepts 1-20 letters/digits/dots/underscores/hyphens with a letter first; prevents malformed usernames entering the filter
if / throw	Checks a condition and stops on invalid input
continue	Skips the current CSV record when the username already exists
$null = query	Discards a successful lookup's display output while allowing errors to surface
$(...) inside a quoted string	Inserts the value of an expression into the output text
$true	Boolean true, used to enable the account and require its password change
try / catch and $_.Exception.Message	Catches account/group operation errors and prints a readable FAILED message

Existing usernames are skipped without password resets or membership changes. Account creation and group assignment are separate operations; a FAILED row can leave a partially completed account and needs inspection before retrying. CREATED is printed only after both operations return successfully.

Run the saved script

Inside DC01's elevated Windows PowerShell, run:

POWERSHELL

```
& "C:\LabScripts\New-LabUsers.ps1"
```

The & call operator runs the script at the quoted path. The script reads C:\LabScripts\NewUsers.csv and applies its AD changes under the signed-in administrator.

Enter a temporary password privately when prompted for each new user. Guidance for this lab is 16 or more characters with upper/lowercase letters, a number, and a symbol, excluding the person's names or username. The actual domain minimum length has not been queried; 16 is guidance, not a measured policy value.

Expected messages on a successful first run:

TEXT

```
CREATED: alex.taylor -> GG_IT_Users
CREATED: maya.patel -> GG_HR_Users
```

An already-existing username produces SKIPPED. A FAILED message needs review; neither it nor SKIPPED proves that provisioning completed. Account properties and group memberships will be queried after the run.

Actual result: The user reported matching both expected CREATED messages: alex.taylor to GG_IT_Users, and maya.patel to GG_HR_Users. Script execution and these success messages are recorded by user report, without a screenshot. Stored account properties and memberships are queried next. No temporary password is recorded in this guide.

28. Verify the CSV-provisioned accounts

Where and shell: Inside DC01, as CORP\Administrator, in Windows PowerShell opened with Run as administrator.

Purpose: Read the saved AD values after provisioning to verify the accounts' enabled state, department, OU location, and direct group membership.

Run one command for each new user:

POWERSHELL

```
Get-ADUser -Identity alex.taylor -Properties Department,MemberOf | Select-Object
  Name,SamAccountName,Enabled,Department,DistinguishedName,MemberOf
Get-ADUser -Identity maya.patel -Properties Department,MemberOf | Select-Object
  Name,SamAccountName,Enabled,Department,DistinguishedName,MemberOf
```

Command part	Meaning and reason
Get-ADUser	Retrieves the account's stored AD properties
-Identity alex.taylor / maya.patel	Selects the corresponding sign-in name
-Properties Department,MemberOf	Requests department and group-membership attributes in addition to default properties
Pipeline	Passes the account object from Get-ADUser to Select-Object
Select-Object	Selects only the listed properties for inspection
Name / SamAccountName	Displays the full object name and short sign-in name
Enabled	Shows whether the account is enabled
Department	Shows IT or HR
DistinguishedName	Shows the directory object's full path, allowing LabUsers OU placement to be checked
MemberOf	Shows direct group-membership distinguished names, allowing the department-group assignment to be checked

Effect: Both commands are read-only; they query AD and select output properties.

Account	Expected Name	Enabled	Department	DistinguishedName must include	MemberOf must include
alex.taylor	Alex Taylor	True	IT	OU=LabUsers	CN=GG_IT_Users
maya.patel	Maya Patel	True	HR	OU=LabUsers	CN=GG_HR_Users

A distinguished name (DN) identifies an object's directory path. For example, Alex's expected path is CN=Alex Taylor,OU=LabUsers,DC=corp,DC=sampath,DC=test. CN means Common Name; OU means Organizational Unit; the DC components in this path mean Domain Component.

Actual result: The user confirmed that both accounts matched the expected values. Alex Taylor / alex.taylor is enabled, in IT and LabUsers, with GG_IT_Users membership. Maya Patel / maya.patel is enabled, in HR and LabUsers, with GG_HR_Users membership. These stored-value checks are recorded by user report; no query-output screenshot was supplied. First-sign-in and resource-access tests follow.

29. Test the scripted user's first sign-in and share access

Where: Inside CLIENT01. Keep DC01 and FS01 running.

Purpose: Test a representative script-created account through first sign-in, the required password change, and actual file access. Alex's account should receive the same IT access pattern already demonstrated with Sampath.

1. Sign out of CLIENT01. Choose Other user and sign in as CORP\alex.taylor using the temporary password entered when the provisioning script ran.
2. When Windows requires a password change, set a new password and finish signing in. Use a different password that meets the domain policy and excludes the person's names. Password values are kept outside this guide.
3. Open a normal Command Prompt through Windows+R and cmd, then run:

CMD

```
whoami
```

Command part	Meaning and reason
whoami	Reads the identity running this Command Prompt; expected output is corp\alex.taylor so the share test is tied to the script-created standard user

Effect: Read-only identity check.

4. Use Windows+R to open each UNC path below while signed in as Alex:

Path	Action and expected result
\\FS01\IT	Opens. Create Alex-Test.txt, enter and save text, reopen and edit/save it, then delete the test file. All operations should succeed.
\\FS01\HR	Access denied or a no-permission message.

The UNC paths select department shares on FS01. File operations change only the disposable test file; the HR check should be denied for this IT user.

Check	Expected	Actual
Alex's first sign-in and required password change	Successful	Passed by user report
whoami on CLIENT01	corp\alex.taylor	Included in procedure; output not separately supplied
IT share file operations	Allowed	Passed by user report
HR share access	Denied	Passed by user report

Actual result: The user confirmed that Alex's password change, IT file operations, and HR denial all worked. These functional outcomes are recorded by user report, without screenshots or separate whoami output. Maya's directory properties and group membership are confirmed by user report, but her first sign-in has not been tested.

30. Practice controlled account disablement and restoration

Purpose: Simulate the AD account-disable part of offboarding with the fictional alex.taylor account, test a fresh sign-in, and restore the account afterward for continued lab use.

Save the disablement script on DC01

Where: Inside DC01, as CORP\Administrator. Open a new Notepad window and save the script as C:\LabScripts\Disable-LabUser.ps1, with Save as type All Files (*.*) and Encoding UTF-8.

POWERSHELL

```
param(
    [Parameter(Mandatory = $true)]
    [string]$Username
)

$ErrorActionPreference = "Stop"
Import-Module ActiveDirectory
$User = Get-ADUser -Identity $Username
$LabOU = "OU=LabUsers,DC=corp,DC=sampath,DC=test"

if ($User.DistinguishedName -notlike "*,$LabOU") {
    throw "The account must be in LabUsers."
}

Disable-ADAccount -Identity $User
Get-ADUser -Identity $User | Select-Object SamAccountName,Enabled
```

Command or construct	Meaning and reason	Effect
param and [string]$Username	Accepts the target username as a script argument	Reads input
[Parameter(Mandatory = $true)]	Requires a username when the script runs	Validates required input
$ErrorActionPreference = Stop	Stops processing if a command fails	Sets script error behavior
Import-Module ActiveDirectory	Loads the AD commands	Changes the PowerShell session

Command or construct	Meaning and reason	Effect
Get-ADUser -Identity $Username	Reads the target account, including its directory path	Read-only AD query
$LabOU	Stores the allowed LabUsers OU's distinguished name	Stores a setting
-notlike with the OU suffix pattern	Rejects an account outside the allowed OU before any disablement	Checks scope
throw	Stops the script when the OU check fails	Stops processing
Disable-ADAccount -Identity $User	Disables the specific account that passed the scope check	Changes AD account state
Get-ADUser -Identity $User	Reads the stored account state again after disablement	Read-only AD query
Pipeline and Select-Object SamAccountName,Enabled	Pass the account object onward and display its username and enabled flag	Selects output properties

Run the script and verify disabled state

Where and shell: Inside DC01, in Windows PowerShell opened with Run as administrator, as CORP\Administrator.

POWERSHELL

```
& "C:\LabScripts\Disable-LabUser.ps1" -Username alex.taylor
```

The & operator runs the script; -Username supplies alex.taylor to its mandatory parameter. Expected output: SamAccountName alex.taylor, Enabled False. Disablement retains the directory account for this lab exercise.

Test a fresh sign-in on CLIENT01

Keep DC01 running and reachable. On CLIENT01, fully sign out of Alex's existing session, then try to sign in again as CORP\alex.taylor with the new password set at first sign-in. Expected result: a message that the account is disabled and sign-in is blocked.

Fully signing out is part of this test: it checks a new domain sign-in rather than relying on an already-open desktop or share session. This exercise does not test termination of existing sessions or ticket revocation.

Restore the account on DC01

After the blocked-sign-in test, return to DC01's elevated Windows PowerShell and run:

POWERSHELL

```
Enable-ADAccount -Identity alex.taylor
Get-ADUser -Identity alex.taylor | Select-Object SamAccountName,Enabled
```

Command part	Meaning and reason
Enable-ADAccount	Re-enables the test account after the exercise
-Identity alex.taylor	Selects Alex's account by its short domain sign-in name
Get-ADUser and Select-Object	Read the resulting username and enabled flag

Effect and expected result: Enable-ADAccount changes the AD account state. The following query is read-only and should show alex.taylor with Enabled True. Then sign in again on CLIENT01 using Alex's current password; sign-in should succeed.

Check	Expected	Actual
Disablement script scope and execution	alex.taylor processed in LabUsers	Completed by user report; script output not separately supplied
AD query after disablement	Enabled False	Passed by user report
Fresh CLIENT01 sign-in while disabled	Blocked as account disabled	Passed by user report
AD query after restoration	Enabled True	Passed by user report
CLIENT01 sign-in after restoration	Successful	Passed by user report

Actual result: The user confirmed the requested sequence: Enabled False, blocked sign-in while disabled, Enabled True after restoration, and successful sign-in afterward. These outcomes are recorded by user report without screenshots. Alex is restored to enabled state for continued lab use. This exercise demonstrates reversible AD account disablement and fresh authentication checks; it does not demonstrate termination of existing sessions or complete organizational offboarding.

Phase 7 - Backup, recovery, and final checkpoint

31. Install Windows Server Backup on FS01

Purpose: Install Windows Server Backup (WSB), then back up sample department files and prove that a deleted file can be restored with its contents intact.

Where and shell: Inside FS01, in Windows PowerShell opened with Run as administrator, using an administrator account. These commands run in the file-server guest, not DC01, CLIENT01, or the host laptop.

POWERSHELL

```
Install-WindowsFeature -Name Windows-Server-Backup -IncludeManagementTools
Get-WindowsFeature -Name Windows-Server-Backup
```

Run the installation command first and wait for it to finish. If its RestartNeeded value is Yes, restart FS01 before running the verification command.

Command or parameter	Meaning and reason	Effect
Install-WindowsFeature	Installs the selected server feature on the local FS01 server	Changes installed Windows components
-Name Windows-Server-Backup	Selects the built-in Windows Server Backup feature	Identifies the feature to install
-IncludeManagementTools	Includes management tools associated with the selected feature	Adds available management components
Get-WindowsFeature	Reads the installation state of the selected feature	Read-only query
-Name Windows-Server-Backup in the query	Limits the result to this backup feature	Selects the result

Expected results: Installation reports Success True. The verification query reports Windows Server Backup with Install State Installed, typically displayed with [X]. Installation alone does not create a backup.

Actual result: The user confirmed that installation showed Success True and the verification query showed Installed. These results are recorded by user report without a screenshot. RestartNeeded was not separately reported. No backup job, backup copy, or recovery has been configured or tested yet.

32. Add FS01's virtual backup disk

Where: VirtualBox Manager on the Windows host laptop. Shut down FS01 through its Windows Start menu first and wait for VirtualBox to show Powered Off.

Purpose: Provide a separate destination disk for the upcoming file-backup exercise. Department files remain the backup source on FS01's existing C: volume. This additional virtual disk is a lab destination on the same laptop; it does not protect against loss of the laptop or its underlying storage.

1. Select FS01 in VirtualBox Manager and open Settings, then Storage.
2. Select Controller: SATA and use Add Disk, then Hard Disk.
3. In the Hard Disk Selector, choose Create and enter the following settings.

Setting	Planned value and reason
Image file type	VDI (VirtualBox Disk Image), the native virtual disk format
Capacity	40 GB, sufficient for this small sample-file backup exercise
Storage allocation	Dynamically allocated; the host image grows as data is written, up to its configured capacity
Host file location	C:\SysAdminLab\FS01\FS01-Backup.vdi
Attachment	A second disk on FS01's existing SATA controller

4. Finish creating the image, select FS01-Backup.vdi in the selector, and click Choose.
5. Inspect the Storage page: FS01 should have its original 80 GB system disk and the added 40 GB backup disk. Save with OK, start FS01, and sign in as an administrator.

Effect: Creating the VDI writes a new disk-image file on the host; attaching it changes FS01's virtual hardware configuration. These instructions do not initialize, partition, or format a guest disk.

Expected result: Two virtual hard disks attached to FS01, with the new image named FS01-Backup.vdi and capacity 40 GB.

Actual result: The creation-dialog screenshot showed 40.00 GB, VDI format, and Pre-allocate Full Size unchecked. The subsequent Hard Disk Selector screenshot, image(20261025-214313).png, showed FS01-Backup.vdi with Virtual Size 40.00 GB and Actual Size 2.00 MB under Not Attached. This verifies that the new image was created, while attachment was still pending at the time of that screenshot. After receiving the Choose, save-settings, start-FS01, and administrator-sign-in instructions, the user confirmed reaching the desktop. Subsequent attachment and boot are therefore recorded by user report; no final Storage-page screenshot was supplied. The host file location was instructed but not independently shown. Those guest-disk details were unverified at this checkpoint; milestones 33-36 record the subsequent disk preparation and backup catalog.

33. Identify FS01's backup disk inside Windows

Where and shell: Inside FS01, in Windows PowerShell opened with Run as administrator, using an administrator account.

POWERSHELL

```
Get-Disk
```

Command	Meaning and reason	Effect
Get-Disk	Lists disks visible to the local Windows guest, including disk numbers, sizes, operational state, and partition styles; used to identify the new 40 GB disk before any disk changes	Read-only inventory

Expected result: Two disks: the existing approximately 80 GB system disk and the new approximately 40 GB disk. The new, uninitialized disk should show Partition Style RAW. In this output, RAW means the disk has no initialized partition table; it is not a filesystem format selected for the backup.

Decision: Use the reported disk number and RAW partition style to target the newly added backup disk. The disk number is established by the user's inventory result, not assumed from the order in which disks were added.

Actual result: The user reported that the new disk is Number 1 with RAW partition style. This identifies the backup disk inside FS01 by user report. The complete command output, including size, operational state, and other disks, was not supplied. The 40 GB capacity is confirmed for the new image by the earlier selector screenshot. The disk was still uninitialized at this check; its subsequent preparation is recorded in milestone 34.

34. Prepare Disk 1 as FS01's backup volume

Where and shell: Inside FS01, in Windows PowerShell opened with Run as administrator, using an administrator account. The target is the newly added, blank 40 GB Disk 1 identified above.

Purpose: Write a GPT partition table, create a backup data partition, and format it as NTFS with the drive letter B: and label LabBackup. This creates a usable storage destination; it does not create a backup.

Run the following steps one at a time. If a command reports an error, stop and inspect that result before entering the next step.

Initialize the new disk

POWERSHELL

```
Initialize-Disk -Number 1 -PartitionStyle GPT -ErrorAction Stop
```

Command or parameter	Meaning and reason	Effect
Initialize-Disk	Prepares a RAW disk for first use by writing a partition table	Changes disk metadata
-Number 1	Selects the new disk number reported from FS01's inventory	Limits the target to Disk 1
-PartitionStyle GPT	Uses the GUID Partition Table layout	Sets the partition-table format
-ErrorAction Stop	Treats a reported command error as terminating for this operation	Stops that failed operation

Expected result: Successful return to the prompt, often with no output because PassThru was not requested. Disk 1's partition style becomes GPT; its backup data partition has not yet been created.

Create and format the backup data partition

POWERSHELL

```
New-Partition -DiskNumber 1 -UseMaximumSize -DriveLetter B -ErrorAction Stop |
    Format-Volume -FileSystem NTFS -NewFileSystemLabel "LabBackup" -ErrorAction Stop
```

Command or parameter	Meaning and reason	Effect
New-Partition	Creates a data partition on the selected disk	Changes the disk's partition layout
-DiskNumber 1	Targets the newly initialized backup disk	Limits where the partition is created
-UseMaximumSize	Uses the maximum space available for this new partition	Sets the data partition's size

Command or parameter	Meaning and reason	Effect
-DriveLetter B	Assigns B: as the backup volume's access letter	Creates the drive-letter assignment
Pipeline operator	Sends the newly created partition object directly to Format-Volume	Keeps formatting tied to that returned partition
Format-Volume	Creates a filesystem on the new partition	Formats the target partition
-FileSystem NTFS	Selects New Technology File System for this Windows backup volume	Sets the filesystem
-NewFileSystemLabel LabBackup	Assigns the readable volume label LabBackup	Labels the formatted volume
-ErrorAction Stop on each cmdlet	Makes a failure terminate the affected operation; if partition creation fails, it supplies no successful new partition for formatting	Error handling

If Windows requests format confirmation, confirm only the newly created backup partition. B: is the chosen backup drive letter; if partition creation reports that the letter is unavailable, stop and resolve the assignment before proceeding. No Force or Clear-Disk operation is used.

Expected result: The format operation returns a volume with drive letter B, label LabBackup, NTFS filesystem, and approximately 40 GB capacity. The filesystem's reported capacity is slightly below the virtual disk capacity because of partition metadata and reserved space.

Verify the resulting volume

POWERSHELL

```
Get-Volume -DriveLetter B
```

Command or parameter	Meaning and reason	Effect
Get-Volume	Reads volume configuration, health, size, and remaining space	Read-only query
-DriveLetter B	Selects the volume assigned the backup letter	Limits the query

Expected result: DriveLetter B, FileSystemLabel LabBackup, FileSystem NTFS, HealthStatus Healthy, and Size approximately 40 GB.

Actual result: The user confirmed that the verification result matched B, LabBackup, NTFS, Healthy, and approximately 40 GB. The preparation procedure and resulting volume are recorded as completed by user report; no output screenshot or separate post-initialization Get-Disk query was supplied. At this stage the backup destination was ready; the first backup follows in milestone 35.

35. Create recovery test data and take the first file backup

Where and shell: Inside FS01, in Windows PowerShell opened with Run as administrator, using an administrator account. All C: and B: paths in this stage refer to drives inside FS01.

Purpose: Back up actual sample data in the department folders to B:, and retain a SHA-256 content fingerprint for the later recovery check. The test file will be deleted and restored only after the backup has completed and its catalog has been checked.

Create and read the sample file

POWERSHELL

```
Set-Content -Path "C:\Shares\IT\Recovery-Test.txt" -Value "SysAdminLab recovery test - original file
  content." -Encoding UTF8 -ErrorAction Stop
Get-Content -Path "C:\Shares\IT\Recovery-Test.txt" -ErrorAction Stop
```

Command or parameter	Meaning and reason	Effect
Set-Content	Creates the named test text file, or replaces its content if the same file already exists	Writes the lab sample file
-Path C:\Shares\IT\Recovery-Test.txt	Places the sample inside the existing IT department folder that will be backed up	Selects the file
-Value	Supplies the original text we expect to recover	Sets test content
-Encoding UTF8	Writes the text using UTF-8 encoding	Controls stored text encoding
Get-Content	Reads the saved text so it can be checked before backup	Read-only file check
-ErrorAction Stop	Makes a reported command error terminate that operation	Error handling

Expected result: Set-Content normally returns no output. Get-Content displays: SysAdminLab recovery test - original file content.

Save a content fingerprint for comparison after recovery

POWERSHELL

```
(Get-FileHash -Path "C:\Shares\IT\Recovery-Test.txt" -Algorithm SHA256 -ErrorAction Stop).Hash |
    Set-Content -Path "C:\Shares\Recovery-Test.sha256.txt" -Encoding ASCII -ErrorAction Stop
```

Command or construct	Meaning and reason	Effect
Get-FileHash -Path	Reads the test file and calculates its content fingerprint	Reads the original file
-Algorithm SHA256	Uses the 256-bit Secure Hash Algorithm for later content comparison	Selects the hash algorithm
Parentheses and .Hash	Complete the calculation, then extract just the hash string	Selects a result property
Pipeline	Sends that hash string to Set-Content	Passes the value onward
Set-Content -Path C:\Shares\Recovery-Test.sha256.txt	Saves the baseline outside the IT test file, so it remains available when that test file is deleted	Writes the hash record
-Encoding ASCII	Saves the hexadecimal hash as plain ASCII text	Controls the record's encoding
-ErrorAction Stop	Terminates the affected operation on a reported error	Error handling

Expected result: No console output. The baseline record contains a 64-character hexadecimal SHA-256 value. It is included in the C:\Shares backup and will remain in place during the later deletion of only the test file. Matching

the recovered file's hash against this saved baseline provides strong evidence that its contents were restored intact.

Start a one-time file backup

Run this only after the sample text check and hash-record command have succeeded:

POWERSHELL

```
wbadmin start backup -backupTarget:B: -include:C:\Shares -vssCopy
```

Command or option	Meaning and reason	Effect
wbadmin	Native Windows backup administration utility, run here from elevated PowerShell	Invokes the backup service
start backup	Starts a one-time backup operation	Creates backup data
-backupTarget:B:	Stores the backup on FS01's prepared LabBackup volume	Selects the destination
-include:C:\Shares	Includes the Shares folder and its contents, including IT, HR, and the baseline record	Selects the source
-vssCopy	Uses Volume Shadow Copy Service copy-backup mode	Specifies the VSS backup type

Review the displayed source and destination; if prompted to start the operation, enter Y and press Enter. Leave FS01 running and wait for the backup to finish. If any command fails, stop and inspect the error before continuing.

Scope: This exercise backs up the selected department files. Full-server, bare-metal, and AD system-state recovery are not configured by this command. The second VDI remains on the same physical laptop, as noted in milestone 32.

Expected result: A successful backup-completion message. Catalog verification and a controlled deleted-file recovery follow after completion is reported.

Actual result: The user reached the native backup-start confirmation prompt, received instructions to enter Y, and subsequently reported being done with the backup. The final backup-job success output was not supplied. The subsequent screenshot in milestone 36 verifies an available backup catalog entry, the original sample content, and the saved hash baseline. Those catalog checks corroborate that a recoverable backup version is recorded; actual file recovery is tested separately.

36. Check the backup catalog and recovery-test prerequisites

Where and shell: Inside FS01, in Windows PowerShell opened with Run as administrator, using an administrator account.

Purpose: Confirm that Windows can find FS01's backup on B:, identify its exact version and supported recovery types, and check the original test data and saved hash before a controlled deletion.

POWERSHELL

```
wbadmin get versions -backupTarget:B: -machine:FS01
```

Command or option	Meaning and reason	Effect
wbadmin get versions	Lists catalog information about backup versions, including time, version identifier, location, and supported recovery types	Read-only catalog query
-backupTarget:B:	Queries the prepared backup destination	Selects the location
-machine:FS01	Selects backups of FS01 at that destination	Selects the backed-up computer

Expected result: A backup entry with its Version identifier and a Can recover entry that supports file recovery. Use the exact identifier shown by Windows for the subsequent recovery command; do not substitute an assumed date or time. Listing the catalog is a prerequisite check and does not itself prove recovery works.

POWERSHELL

```
Get-Content -Path "C:\Shares\IT\Recovery-Test.txt","C:\Shares\Recovery-Test.sha256.txt" -ErrorAction
  Stop
```

Command or parameter	Meaning and reason	Effect
Get-Content	Reads the original sample file and saved baseline record	Read-only file query
-Path with two comma-separated paths	Selects both files, in the order shown	Reads each listed file
-ErrorAction Stop	Stops this query on a reported error, such as a missing prerequisite file	Error handling

Expected result: The original text, SysAdminLab recovery test - original file content., followed by the saved 64-character hexadecimal hash. A missing file, an error, or an unexpected backup catalog result needs investigation before the deletion-and-restore exercise.

Actual result: Screenshot image(20261025-221003).png verifies the following catalog and file-check output:

Verified item	Observed value
Backup time, as displayed by the guest	10/25/2026 3:05 PM
Backup target	Fixed Disk labeled LabBackup(B:)
Version identifier	10/25/2026-22:05
Can recover	Volume(s), File(s)
Original test-file text	SysAdminLab recovery test - original file content.
Saved SHA-256 baseline	833170925B8F7BEF1A0CE86F8B60313021D847AB1BCAD2550D3B65B1E9701223

The exact version identifier shown by the catalog is retained for recovery. This screenshot confirms that the original file and baseline are present and that file recovery is offered; it does not yet demonstrate a successful restoration.

37. Recover the deleted test file and verify its contents

Where and shell: Inside FS01, in Windows PowerShell opened with Run as administrator, using an administrator account. The synthetic test file is C:\Shares\IT\Recovery-Test.txt. Its backup and saved baseline were checked in milestone 36.

Purpose: Simulate deletion of the sample file, restore it from the verified backup version, then compare the recovered text and SHA-256 hash with the original baseline.

Delete the sample file and confirm it is absent

POWERSHELL

```
Remove-Item -LiteralPath "C:\Shares\IT\Recovery-Test.txt" -ErrorAction Stop
Test-Path -LiteralPath "C:\Shares\IT\Recovery-Test.txt"
```

Command or parameter	Meaning and reason	Effect
Remove-Item	Deletes the synthetic test file to create the recovery scenario	Deletes the selected file
-LiteralPath	Uses the exact file path without interpreting wildcard characters	Limits the deletion to the named test file
-ErrorAction Stop	Terminates the operation on a reported deletion error	Error handling
Test-Path -LiteralPath	Checks whether the named file exists afterward	Read-only existence check

Expected result: Deletion normally produces no output. Test-Path returns False. Continue to recovery after this absence check succeeds.

Restore from the selected backup

POWERSHELL

```
wbadmin start recovery -version:10/25/2026-22:05 -itemtype:File -items:C:\Shares\IT\Recovery-Test.txt
  -backupTarget:B: -machine:FS01
```

Command or option	Meaning and reason	Effect
wbadmin start recovery	Starts recovery using the chosen backup version	Restores backed-up data
-version:10/25/2026-22:05	Selects the exact version verified in the catalog screenshot	Chooses the recovery point
-itemtype:File	Requests file recovery	Selects the recovery type
-items:C:\Shares\IT\Recovery-Test.txt	Identifies the individual sample file in that backup	Limits the recovery item
-backupTarget:B:	Reads the backup from LabBackup	Selects the backup location
-machine:FS01	Identifies the backed-up computer at that location	Selects the computer

The command uses the original file location, with no alternate recovery target. Review the displayed recovery details and enter Y if asked to start. Wait for successful completion before the next check; inspect any error before proceeding.

Read the recovered text and compare hashes

POWERSHELL

```
Get-Content -Path "C:\Shares\IT\Recovery-Test.txt" -ErrorAction Stop
$ExpectedHash = Get-Content -Path "C:\Shares\Recovery-Test.sha256.txt" -ErrorAction Stop
$RestoredFile = Get-FileHash -Path "C:\Shares\IT\Recovery-Test.txt" -Algorithm SHA256 -ErrorAction
  Stop
$RestoredFile.Hash -eq $ExpectedHash
```

Command or construct	Meaning and reason	Effect
Get-Content on Recovery-Test.txt	Displays the recovered sentence	Read-only content check
$ExpectedHash = Get-Content on Recovery-Test.sha256.txt	Reads the pre-backup baseline and stores it for comparison	Reads a file and assigns a session variable
$RestoredFile = Get-FileHash	Calculates the recovered file's SHA-256 result and stores the returned object	Reads the restored file and assigns a session variable
-Algorithm SHA256	Uses the same hash algorithm as the original baseline	Selects the hash algorithm
.Hash	Extracts the hash string from the returned hash object	Reads an object property
-eq	Compares the recovered hash with the saved original hash	Returns a Boolean comparison result
-ErrorAction Stop	Stops a failed read or calculation, such as a missing file	Error handling

Expected result: Get-Content displays the original sentence. The final comparison returns True, giving strong evidence that the recovered file's bytes match the saved original baseline. This checks content integrity; milestone 38 records the subsequent standard-user access test.

Recovery check	Expected	Actual
Test file absent after controlled deletion	Test-Path False	Procedure completion reported; exact Boolean not separately supplied
Recovery operation	Successful completion	Completed by user report
Recovered text	Original sentence	Confirmed by user report through the CLIENT01 access check in milestone 38
Restored SHA-256 vs. original baseline	True	Explicitly confirmed by user report

Actual result: The user reported completing the deletion/recovery procedure and subsequently explicitly confirmed that the final FS01 hash comparison returned True. The recovered original text was also confirmed through the standard-user check in milestone 38. Content recovery and the matching hash are recorded by user report; the deletion Boolean and recovery-job output were not separately supplied as screenshots.

38. Verify restored-file access from CLIENT01

Where: Inside CLIENT01. Keep DC01 and FS01 running. Sign in as the standard domain account CORP\alex.taylor.

Purpose: Check that an authorized IT user can read the recovered file through the SMB share, and confirm the final hash result from FS01. Opening the file as an administrator alone does not establish the standard user's effective access.

Open a normal Command Prompt and run:

CMD

```
whoami
```

Command	Meaning and reason	Effect
whoami	Shows the current sign-in identity; expected corp\alex.taylor ties the share check to the standard IT test user	Read-only identity query

Use Windows+R to open these UNC paths:

Path	Expected result and purpose
\\FS01\IT\Recovery-Test.txt	Opens and shows SysAdminLab recovery test - original file content.; demonstrates standard-user read access to the recovered file
\\FS01\HR	Access denied; checks that the IT user's department restriction still applies

Close the recovered text file after reading it. Confirm whether the final hash comparison already run on FS01 returned True. These checks read the recovered file and its access behavior; they do not change its content.

Actual result: The user explicitly confirmed all three requested outcomes: the recovered file opens for Alex with its original text, HR access is denied, and the earlier FS01 hash comparison returned True. These are passed by user report. The whoami query was included in the procedure; its exact output was not separately supplied. The completed checks demonstrate restored-file read access for the IT user, continued department separation, and a content hash matching the original baseline.

39. Save the post-recovery lab checkpoint

Where: Shut down each guest through its own Windows Start menu, then use VirtualBox Manager on the host laptop.

Purpose: Preserve the working domain, DHCP, file shares, account automation state, and verified recovery result before further changes. A VirtualBox snapshot captures that VM's virtual disk state and settings at the checkpoint.

1. Shut down CLIENT01, then FS01, then DC01 through Windows.
2. Wait until all three VMs show Powered Off in VirtualBox Manager.
3. For each VM, open Snapshots and select Take. Use the names below.

VM	New snapshot name
DC01	DC01-Backup-Recovery
FS01	FS01-Backup-Recovery
CLIENT01	CLIENT01-Backup-Recovery

4. Confirm all three named snapshots are listed. Leave the VMs powered off while the project guide is consolidated.

Effect: Writes new VirtualBox snapshot metadata and disk checkpoint state. No snapshot restoration is requested in this step. The separate file backup already created on B: remains part of the recovery exercise.

Expected result: Each VM lists its named post-recovery snapshot, with the three guests powered off.

Actual result: The user confirmed that all three named post-recovery snapshots were saved after the shutdown procedure. Snapshot creation and the powered-off checkpoint are recorded by user report; final snapshot-list screenshots were not supplied. The earlier file-sharing snapshots remain historically unconfirmed. No restoration of the complete three-VM snapshot set was tested.

Operating the lab

Task	Procedure and reason
Start the lab	Start DC01 first and allow domain services to become ready; then start FS01 and CLIENT01
Shut down the lab	Use Windows shutdown on CLIENT01, then FS01, then DC01
Review commands	Use the milestone for the intended VM; the command tables distinguish read-only checks from changes
Use standard accounts	Use a department user for share-access checks so administrator rights do not obscure the intended permissions
Use scripts	NewUsers.csv, New-LabUsers.ps1, and Disable-LabUser.ps1 belong in DC01's C:\LabScripts; their full source and use are in milestones 26-30
Review file recovery	The known recovery point and the SHA-256 comparison procedure are in milestones 36-38
Review saved state	The three post-recovery snapshots are recorded in milestone 39; the guests were shut down for that checkpoint

Troubleshooting and useful lessons

Situation	Explanation and resolution used
Ctrl+Alt+Delete affected the host	Send it through the VM's Input > Keyboard menu, or use the VirtualBox host-key shortcut described during setup
Copy and paste did not work in a VM	Install Guest Additions inside the guest, restart it, and enable the relevant Shared Clipboard direction; the completed setup step was user-reported, while functionality in every guest was not independently demonstrated
Windows installation ISO occupied the virtual CD drive	Eject that ISO before mounting the Guest Additions CD image
DNS delegation warning during promotion	The new private test domain has no authoritative parent DNS zone to update; the wizard's prerequisites passed and domain creation succeeded
Promotion restarted automatically	The subsequent CORP administrator sign-in and AD checks established that promotion completed
A department user received Access denied	Denial for the other department is the intended result; use the correct standard account and refresh sign-in after group membership changes
Role installation reported no restart needed	A successful install with RestartNeeded No does not require an extra restart for that role-install step
Newly added backup disk showed RAW	It needed initialization and a partition/filesystem before use; the identified Disk 1 became B: LabBackup through milestone 34
Backup asked for confirmation	The selected source and target were reviewed, then Y started the one-time backup
Secure-channel checks	Test-ComputerSecureChannel was used on domain members; DC01's health was checked with domain-controller and DNS checks instead

The guide records useful resolutions and final configurations rather than every routine click, delay, or typing correction.

Resume project description

Windows Server and Active Directory Home Lab | Personal Project

- Built a three-VM lab with Active Directory Domain Services, DNS, Windows DHCP, domain joining, and Group Policy.
- Secured IT and HR SMB shares with NTFS permissions and department security groups; automated two domain users from CSV with PowerShell and tested account disablement/restoration.
- Backed up department files, restored a deleted test file, confirmed its SHA-256 hash matched the original, and verified access through a standard domain account.

Present this as a home lab project. The recorded tests support these statements; no production employment, certification, cloud deployment, or measured recovery-time claim is implied.

Scope and follow-up work

The core lab is complete through the reported three-VM post-recovery checkpoint. The build and command record, configuration tables, topology, results, and glossary have been consolidated into this guide.

Follow-up or limit	Current status
Supported client migration	Windows 11 Pro migration remains planned. The current guest is Windows 10 Pro; no upgrade or Extended Security Updates enrollment was verified
Backup coverage	A one-time file backup and single-file recovery were exercised; full-server, system-state, scheduled, and off-host backup/recovery were not implemented
Physical failure protection	Both virtual disks are on the same laptop; the added backup VDI does not protect against loss of that physical storage
Maya's functional tests	AD properties and group membership were checked; first sign-in and resource access were not tested
Historical file-sharing snapshots	Creation of the three earlier FileShares snapshots remains unconfirmed; the later Backup-Recovery snapshots were confirmed by user report
Complete VM-set restoration	Snapshot creation was reported; a coordinated restoration of all three VMs was not tested
Timing and scale	No measured recovery duration, bulk-user scale beyond the two CSV accounts, or production load test was recorded
Additional projects	Linux administration, monitoring, cloud infrastructure, and certification study are separate next projects

Abbreviations and key terms

Term	Full form or meaning	Use in this project
VM	Virtual Machine	Software-created computer such as DC01
WSB	Windows Server Backup	Feature installed on FS01 for the backup and recovery exercise; installation confirmed by user report
OS	Operating System	Windows Server manages the guest computer
ISO	Optical disk image format, associated with ISO 9660	Installation media stored as a file
CPU	Central Processing Unit	Processes instructions
vCPU	Virtual Central Processing Unit	Processing capacity presented to the VM
RAM	Random Access Memory	Temporary working memory
MB / GB	Megabyte / Gigabyte	Units used for memory and storage settings
VDI	VirtualBox Disk Image	The VM's virtual hard-disk file
SATA	Serial Advanced Technology Attachment (Serial ATA)	Virtual storage-controller interface used to attach FS01's additional backup disk
GUI	Graphical User Interface	Visual management windows and controls
EFI	Extensible Firmware Interface	Firmware interface used during startup
NAT	Network Address Translation	Translates addresses for traffic between networks
NAT Network	A shared VirtualBox network mode	Connects lab VMs and provides an outbound NAT route
NIC	Network Interface Card	The VM's virtual network adapter
IP	Internet Protocol	Network addressing and routing
IPv4 / IPv6	Internet Protocol version 4 / version 6	Addressing protocols; this lab is configured using IPv4
CIDR	Classless Inter-Domain Routing	Prefix notation such as /24
DHCP	Dynamic Host Configuration Protocol	Automatic address and network-setting assignment

Term	Full form or meaning	Use in this project
DHCP scope	Address pool and subnet configuration on a DHCP server	Client pool: 10.10.10.100 through 10.10.10.199
Lease	Temporary assignment of an IP address to a client	CLIENT01's screenshot shows 10.10.10.100 leased from DC01
DHCP options	Additional network settings delivered with a lease	Router, DNS server, and DNS domain in this lab
Scope ID	Identifier of a DHCP scope, expressed as its subnet address	10.10.10.0 identifies the lab scope
DNS	Domain Name System	Name and service-location resolution
A record	DNS address record	Maps a name such as DC01.corp.sampath.test to an IPv4 address
SRV	Service record	Publishes a service's hostname and port so clients can find it
LDAP	Lightweight Directory Access Protocol	The service-discovery record in this lab identifies DC01 on port 389
DFSR	Distributed File System Replication	Service used for SYSVOL replication between domain controllers
NETLOGON	Domain-controller share name	Publishes the domain's sign-in scripts folder
Loopback	Address referring to the same computer	127.0.0.1 lets DC01 use its own DNS service
AD	Active Directory	Microsoft's directory technologies; here, the domain directory context
DS in AD DS	Domain Services	The last two words of the Windows role name
AD DS	Active Directory Domain Services	Central domain directory and identity services
DC	Domain Controller	Server configured to provide AD DS for a domain
TCP	Transmission Control Protocol	Transport protocol named in the TCP/IPv4 settings
ICMP	Internet Control Message Protocol	Used by ping for echo requests and replies
LAN	Local Area Network	A locally grouped network of devices

Term	Full form or meaning	Use in this project
RSAT	Remote Server Administration Tools	Tools used to manage Windows Server roles
PowerShell	Windows command shell and scripting environment	Administrative commands and later automation
Cmdlet	A PowerShell command	For example, Get-WindowsFeature
Parameter	An option or argument controlling a command	-Port 389 chooses the destination TCP port for Test-NetConnection
Pipeline	Passing one command's output objects to another command	Sends the domain password-policy object to Select-Object
Native Windows utility	Windows program that can run in a command shell	hostname, net share, and gpupdate are examples
Command Prompt / CMD	Windows command shell	Used for the sign-in verification and computer-policy refresh
Port	Number identifying a service endpoint	TCP 389 is used for the LDAP connection test
Host / Guest	Physical virtualization computer / OS inside a VM	Laptop / Windows Server in DC01
Hypervisor	Software that creates and runs VMs	Oracle VirtualBox
Static IP	Manually configured IP address	Keeps DC01 at a consistent address
Gateway	Router used for other networks	10.10.10.1 in this lab
Snapshot	VM rollback checkpoint	DC01-Clean-Updated preserves the baseline
Promotion	Configuring a server as a domain controller	Completed for DC01
Domain	Managed identity and policy environment	corp.sampath.test contains the lab's directory objects
Forest	Top-level Active Directory structure	Created with corp.sampath.test as its first domain
FQDN	Fully Qualified Domain Name	Complete DNS name of a computer or domain
NetBIOS	Network Basic Input/Output System	CORP is the short domain name used in CORP\Administrator
GC	Global Catalog	Directory search service selected for the first domain controller

Term	Full form or meaning	Use in this project
RODC	Read-only Domain Controller	Not selected; DC01 is a writable domain controller
Functional level	AD capability level based on supported domain-controller versions	Windows Server 2025 selected for this new forest and domain
DSRM	Directory Services Restore Mode	Domain-controller recovery mode; password set and kept privately
SYSVOL	System Volume	Shared Group Policy files and scripts on a domain controller
NTDS	NT Directory Services	Name used for the AD DS database folder and directory service
OU	Organizational Unit	Organizes domain objects
DN	Distinguished Name	Full AD object path used to verify the account's location
CN in a DN	Common Name	Identifies an object such as CN=Alex Taylor or CN=GG_IT_Users
DC in a DN	Domain Component	DC=corp,DC=sampath,DC=test represents the domain's name components
ADUC	Active Directory Users and Computers	Console used to organize LabUsers, LabComputers, and their accounts
UPN	User Principal Name	Full sign-in name such as sampath@corp.sampath.test
Local account	User account stored on one computer	LabAdmin belongs to CLIENT01
Domain account	User account managed by AD DS	CORP\sampath can authenticate through the domain
Computer account	Directory identity for a domain member	Created for CLIENT01 when it joins the domain
Domain join	Registering a computer as a domain member	Connects CLIENT01 to corp.sampath.test
LOGONSERVER	Windows environment variable identifying the session's logon server	Expected value \\DC01 in the client session
GPO	Group Policy Object	LabComputers-LogonMessage applies the workstation sign-in message

Term	Full form or meaning	Use in this project
GPMC	Group Policy Management Console	Creates and links the domain GPO; its editor configures the settings
Policy link	Association between a GPO and a scope such as an OU	Links LabComputers-LogonMessage to LabComputers
Computer Configuration	GPO settings processed for the computer	Holds the sign-in-message settings
gpupdate	Windows Group Policy refresh command	Reapplies computer settings on CLIENT01
Elevation	Running a process with administrator privileges	Used for the client computer-policy refresh
GG	Global Group; the lab's group-name prefix	GG_IT_Users and GG_HR_Users organize department membership
IT	Information Technology	Department represented by GG_IT_Users
HR	Human Resources	Department represented by GG_HR_Users
Security group	Collection of accounts that can be assigned resource permissions	Used for the coming department-folder access exercise
Global group	Group scope whose account members come from the same domain	Selected for the department membership groups
FS	File Server; the role indicated by the name FS01	Domain member being prepared to host department folders
SMB	Server Message Block	Protocol used to access file shares from CLIENT01
Secure channel	Trust communication between a member computer and its domain	Test-ComputerSecureChannel returned True on FS01 per user report
ACL	Access Control List	Defines access permissions
UNC	Universal Naming Convention	Network path syntax such as \\FS01\IT, identifying a server and share
DACL	Discretionary Access Control List	The part of a security descriptor that lists which identities are allowed or denied access; managed here with icacls

Term	Full form or meaning	Use in this project
ACE	Access Control Entry	One permission entry in an access control list
NTFS	New Technology File System	Windows filesystem used for department permissions and FS01's B: backup volume
GPT	GUID Partition Table	Partition-table layout used in Disk 1's preparation procedure, completed by user report
GUID	Globally Unique Identifier	The identifier type named in GUID Partition Table
RAW partition style	Disk state with no initialized partition table	User-reported initial state of FS01's new Disk 1; RAW is not a chosen filesystem here
VSS	Volume Shadow Copy Service	Windows service used by the copy-backup procedure
Backup catalog	Record of available backup versions and recovery information	Queried to find FS01's exact recovery version on B:
SHA-256	Secure Hash Algorithm with a 256-bit result	Selected for the original and recovered test-file content comparison
Hash	Content fingerprint calculated from file bytes	Saved before backup for a later recovery comparison
UTF-8	8-bit Unicode Transformation Format	Text encoding selected for the recovery test file
ASCII	American Standard Code for Information Interchange	Plain-text encoding selected for the saved hexadecimal hash
CSV	Comma-Separated Values	Input file for planned Alex Taylor and Maya Patel account provisioning
Offboarding	Process of removing access when a person leaves	This lab simulates the AD account-disable part using Alex, then restores the test account

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